

ION DYNAMICS IN THE VENUS IONOSPHERE

K. L. MILLER

Center for Atmospheric and Space Sciences, Utah State University, Logan, UT 84322, U.S.A.

and

R. C. WHITTEN

Space Science Division, NASA Ames Research Center, U.S.A.

Abstract. Dynamics play an important role in defining the characteristics of the Venus ionosphere. The absence of a significant internal magnetic field at Venus allows the ionization to respond freely to gradients in the plasma pressure. The primary response to a gradient in plasma pressure is the nightward flow of the ionization away from a photoionization source on the dayside. The flow is approximately symmetric about the Sun–Venus axis and provides the source of O^+ that maintains the nightside ionosphere during solar maximum. Modelling efforts have generally been successful in describing the average nightward ion velocity. Asymmetric and temporally-variable flow is measured, but is not well described by the models. Departures from axially-symmetric flow described in this paper include ionospheric superrotation at low altitudes and an enhanced flow at high altitude at the dawn terminator. Variability that is the result of changes in the ionopause height induced by changes in solar wind dynamic pressure is especially strong on the nightside. Ion flow to the nightside is also reduced during solar minimum because of a depressed ionopause.

1. Introduction

Beginning with Mariner 2 in 1962, 19 missions have been launched to study Venus (Colin, 1983; Mahajan and Kar, 1988). Until Venera 9 and 10 in 1975, all were fly-bys. The later Venera spacecraft orbited at altitudes above the ionosphere or probed the lower atmosphere. Most carried probes that descended through the atmosphere to the surface. Much was learned about the ionosphere from radio occultation experiments on the early probes. But until the *in situ* measurements of Pioneer Venus, the dynamics of the ionosphere could only be inferred from the shapes of the radio occultation profiles.

Since its insertion into orbit in December 1978, Pioneer Venus has provided an unprecedented opportunity to study the dynamics of the ionosphere of a planet with a substantial ionosphere, but with no strong internal magnetic field. Analysis of the data from the complement of instruments on board the Pioneer–Venus orbiter has broadened our understanding of the plasma processes in the ionosphere of Venus and of the effects of solar radiation and solar wind. Increased availability of data has led to theoretical studies of the physical and chemical properties of the ionosphere. It is evident from these studies that ion dynamics play a key role in defining the characteristics of the ionosphere, especially on the nightside of the planet.

This paper presents a summary of measurements of the ion velocity in the ionosphere of Venus, and a discussion of the mathematical theories that help us understand the measurements. The majority of the ion velocity measurements presented here made by the Retarding Potential Analyzer (ORPA) on board the Pioneer–Venus orbiter space-

craft. Inferences of superthermal ion flow have also been made by the ion mass spectrometer (OIMS).

Pioneer Venus was inserted into orbit around Venus in December 1978. The periapsis of the orbiter was maintained near 150 km altitude during the first 600 orbits allowing measurements throughout the topside ionosphere. Ionospheric measurements were also made on several later orbits, during the period while periapsis altitude was rising but below the ionopause, and on the nightside, when the ionosphere extended to altitudes above periapsis. Nearly all of the information on ion dynamics at Venus comes from the first 3.5 Venus years of the Pioneer–Venus mission. Since the Pioneer–Venus orbiter is in the ionosphere only near periapsis, the primary data set includes measurements during three dayside and four nightside seasons. The rising periapsis meant that the dayside ionosphere was not sampled after this time. Ion velocity measurements made on the nightside after the fourth season were made only sporadically.

Data from Venus have shown that its ionosphere is strongly influenced by dynamic processes. There is no strong internal magnetic field to constrain the motion of the ionospheric plasma, as there is on Earth. The ionosphere is thus able to respond freely to internal pressure gradients which cause the ionization to flow across the terminator into the nightside.

The absence of a strong planetary magnetic field and the lack of planetary-scale ionospheric rotation lead to very different conditions within the ionosphere of Venus than experienced on the Earth. Large day-night differences are observed in the ionospheric quantities. The influence of the solar wind is strong at the upper boundary of the ionosphere. Characteristics of the ionosphere itself are dominated by solar electromagnetic radiation on the dayside, and its absence on the nightside. As a result, most of the measured quantities exhibit near-axial symmetry about the Sun–Venus axis. Statistical treatments of ionospheric densities by Miller *et al.* (1980, 1984) show the ion density at low altitudes to be characterized by three regions (Figure 1). The ion dynamics in each of these regions also have unique characteristics. The dayside, extending to a solar zenith angle of 80° , is characterized by an ion density which is controlled by photoionization, and decreases monotonically with increasing solar zenith angle. In this region, the ions are accelerated toward the nightside to speeds approaching the thermal velocity. Nightward of approximately 110° , the ion density is nearly constant with solar zenith angle. Here the supersonic nightward flow slows and becomes chaotic. Separating these region is a 30° -wide region at the terminator where the ion density drops by about an order of magnitude and where ion velocities are greatest.

The ionization source on the dayside is provided by solar photoionization. The ion density near the ionization peak varies with solar zenith angle in the manner expected for local production in a Chapman layer (Cravens *et al.*, 1981). That is, the ion density is, to a first approximation, inversely proportional to the square root of the solar zenith angle. The ion density at higher altitudes is controlled primarily by diffusion (Nagy *et al.*, 1980). On the nightside, however, solar photoionization does not contribute locally to the ionization, and the maintenance of the ionosphere at the observed level requires another source of ionization. In addition, the median ion density being nearly constant

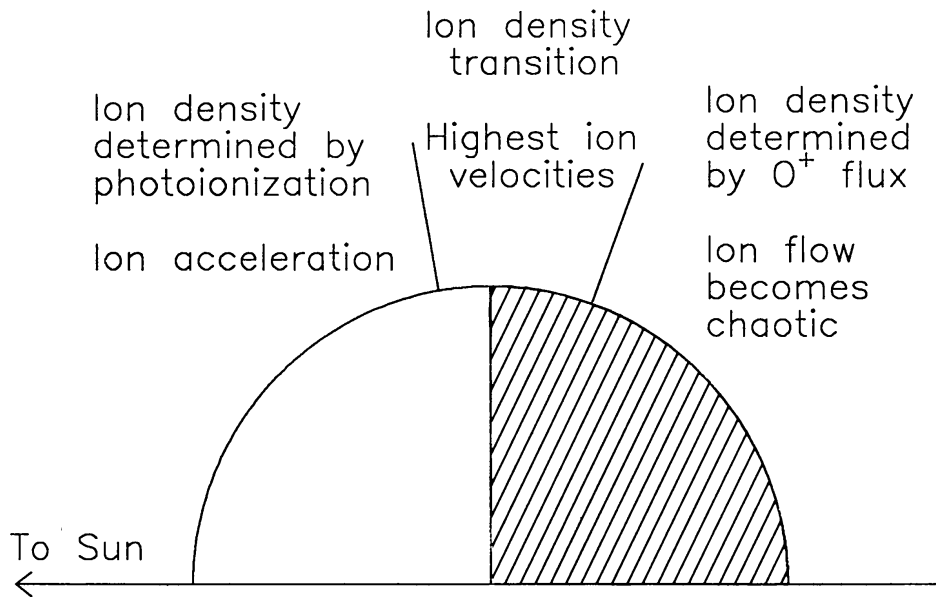


Fig. 1. Approximate location of three regions with similar characteristics of ion density and velocity.

with solar zenith angle requires this source to be acting nearly uniformly on the average throughout the nightside.

The thermal plasma is accelerated to super-sonic velocities as it crosses the terminator region. The flow then slows as it converges on the nightside. The flow of the ionospheric plasma abruptly becomes chaotic at what may be the signature of a recompression shock in the vicinity of 140° solar zenith angle. Although the mean flow is approximately axially-symmetric about the Sun–Venus axis, asymmetries exist between the dawn and dusk hemispheres which influence the characteristics of the ionosphere.

As with many other ionospheric properties that have been measured at Venus, a dominant characteristic of ion velocity in the ionosphere of Venus is variability. There are significant variations in the measured velocity on a time-scale less than the orbital period of the spacecraft, and even within the tens of minutes that Pioneer Venus is in the ionosphere near periapsis. This variability is reflected in the variability of ion density, especially on the nightside of Venus.

The first objective of this paper is to describe the physical relationships that can be formulated mathematically to describe the ion dynamics. Second, the mean characteristics of ion flow in the Venus ionosphere will be described. This mean state as determined by observations may not necessarily correspond to an average of perturbed states of the ionosphere. It is, however, valuable in that it represents conditions that are most likely to occur in the ionosphere. Third will be a description of the modelling studies that have addressed the question of Venus ion dynamics. Fourth, some of the main asymmetries and deviations from the mean flow will be described. This includes the observed dawn-dusk asymmetry in the average ion velocity, and the temporal variability that results from changes in solar activity and solar wind pressure.

2. Theory

2.1. CONSERVATIONS EQUATIONS

In treating ionospheric dynamics, it is necessary to investigate the physical laws which govern the production, loss, heating, and transport of ions and electrons; that is, the laws of momentum, energy and mass conservation, of which Equations (1)–(3) are mathematical statements for steady-state conditions.

Momentum

$$\begin{aligned} \rho_i \mathbf{v}_i \cdot \nabla \mathbf{v}_i + \rho_i \mathbf{v}_i \mathbf{v}_i + 2\rho_i \boldsymbol{\Omega} \times \mathbf{v}_i + \nabla \cdot \eta_i \nabla \mathbf{v}_i + \\ + N_i e \mathbf{E} + \kappa \alpha_i N_i \nabla T_i + \mathbf{g} \rho_i = -\nabla p_i. \end{aligned} \quad (1)$$

Energy

$$\nabla \cdot \kappa \nabla T_i - C_{vi} [\nabla \cdot (\mathbf{v}_i T_i) - \frac{T_i}{3} \nabla \cdot \mathbf{v}_i] = -Q_i + L_i. \quad (2)$$

Mass

$$\nabla \cdot \Phi_i = q_i - l_i \quad (3)$$

with the flux, Φ_i , given by

$$\Phi_i = N_i \mathbf{v}. \quad (4)$$

Because $\mathbf{v}$ is the solution of Equation (1), diffusion has been automatically included in the definition of the flux. Simple in appearance, the solution of these equations can, in general, be obtained only by numerical integration and even then only with many simplifying approximations. For example, because of the very small mass, we can simplify Equation (1) for electrons to

$$\nabla p_e = -e \mathbf{E} N_e. \quad (5)$$

The symbols contained in Equations (1)–(3) have the following meanings:

- ρ_i = mass density of the i th charged particle species;
- $\mathbf{v}_i$ = bulk flow velocity of the i th species;
- ν_i = collision frequency of the i th species;
- $\boldsymbol{\Omega}$ = planetary angular velocity;
- η_i = dynamic viscosity of the i th species;
- p_i = gas kinetic pressure of the i th species;
- N_i = concentration of the i th species;
- T_i = gas kinetic temperature of the i th species;
- m_i = mass of the i th ionic species;
- k = Boltzmann constant;
- g = gravitational acceleration (positive sense if downward);
- e = electronic charge;

- $\mathbf{E}$ = electric field, including but not limited to, charge separation electric fields;
 α_i = thermal diffusion coefficient (e.g., see Schunk and Walker, 1969; Nakada and Sullivan, 1980);
 κ = thermal conductivity (see Merritt and Thompson, 1980; Whitten and Knudsen, 1980);
 C_{v_i} = specific heat capacity at constant volume of the i th species;
 Q_i = heating rate;
 L_i = rate of energy loss;
 D_i = diffusion coefficient;
 $\hat{\mathbf{r}}$ = (inward) radial unit vector;
 $\hat{\boldsymbol{\theta}}$ = tangential unit vector;
 H_{p_i} = plasma scale height (i th species);
 q_i = production rate of the i th ionic species;
 l_i = loss rate of the i th ionic species.

Equations (1)–(3) contain within themselves a formulation of an important mechanism for ion acceleration and flow from the dayside to the nightside: that due to the pressure gradient. Neglecting thermal diffusion and assuming the same flow velocities for all ions (that of the major ion species) and the approximate equality of the number density of the major ion species to the electron density, Equation (1) for the major ion species can be written in two dimensions (we assume axial symmetry about the sub-solar–anti-solar axis) as

$$\frac{1}{R} \frac{\partial u^2/2}{\partial \Theta} + w \frac{\partial u}{\partial z} + \frac{wu}{R} + vu + \frac{1}{\rho R} \frac{\partial}{\partial z} \eta \frac{\partial u}{\partial z} = - \frac{1}{\rho R} \frac{\partial p}{\partial \Theta}, \quad (6)$$

$$\frac{\partial w^2/2}{\partial z} + \frac{u}{R} \frac{\partial w}{\partial \Theta} - \frac{u^2}{R} + vw + \frac{1}{\rho R} \frac{\partial}{\partial z} \eta \frac{\partial w}{\partial z} = - \frac{1}{\rho R} \frac{\partial p}{\partial z}, \quad (7)$$

where u and w are, respectively, the horizontal and vertical components of velocity, p_e is the electron gas pressure, η is the coefficient of viscosity given by $\eta = \eta_0 T^{5/2}$ (Braginskii, 1958), and Θ is the solar zenith angle. This equation would hold, approximately, for a non-rotating planet like Venus. The coupled mass and momentum equations, Equations (3), (6), and (7), have been numerically integrated by Whitten *et al.* (1984) (also see McCormick *et al.*, 1987) to yield ion velocities. These equations proved to be very difficult to integrate because of computational instabilities. To render the solution more tractable, although simultaneously considerably less accurate than that obtained by Whitten and coworkers, Singhal and Whitten (1987) ignored the terms in Equation (6) containing the vertical velocity component, w . The horizontal was then easily computed from the modified form

$$\frac{\partial u^2/2}{\partial \mu} = -R(1 - \mu^2)^{-1/2} \left[vu + \frac{1}{\rho} \frac{\partial}{\partial z} \eta \frac{\partial u}{\partial z} \right] - \frac{1}{\rho} \frac{\partial}{\partial \mu} (p_e + p_i), \quad (8)$$

where $\mu = \cos \Theta$. To crudely simulate the large-SZA attenuation, Singhal and Whitten

(1987) multiplied the velocity obtained from Equation (8) by a factor $\sin \Theta$ or equivalently $(1 - \mu^2)^{1/2}$, which provides a crude approximation to the u 's computed from the two-dimensional model of Whitten *et al.* (1984). If we now ignore viscosity and assume that the plasma temperature is only weakly dependent upon SZA, Equation (8) can be integrated to yield (see Singhal and Whitten, 1987, for a derivation and discussion)

$$v_i = (1 - \mu^2)^{1/2} \left\{ \frac{2k}{m} [T_p(1) \ln N_i(1) - T_p(\mu) \ln N_i(\mu)] - 2R \int_0^{\text{SZA}} v_i v \, d\Theta \right\}^{1/2}, \quad (9)$$

where T_p is the plasma temperature (the sum of the ion and electron temperatures) and $\mu = \cos(\text{SZA})$. Because the computed bulk velocity acquires no vertical component when approximation (8) is used, it is not attenuated at large SZA. As a consequence, it shows a maximum value at the anti-solar point, a physical impossibility. An approximate resolution of this problem, obtained by introducing a factor $\sin \Theta = (1 - \mu^2)$ in Equation (9), is justified in Section 3.1.

In the application of the momentum equation (Equation (1)), the inertial acceleration $\mathbf{v} \cdot \nabla \mathbf{v}$ needs to be resolved into its components in spherical geometry. Since we consider in this paper only the horizontal components of the velocity, v_i , the inertial acceleration is approximated by

$$\mathbf{v} \cdot \nabla \mathbf{v} = \hat{\Theta} \frac{v}{R} \frac{\partial v}{\partial \Theta} + \hat{\mathbf{r}} \frac{v^2}{R}, \quad (10)$$

where $\hat{\Theta}$ is a unit vector in the horizontal direction. Only the first term on the right-hand side of Equation (10) is included in computing v_i ; the second term may be significant in increasing the plasma scale height, H_p . The inverse plasma scale height then takes the form

$$\frac{1}{H'_{p_i}} = \frac{1}{H_{p_i}} \left(1 - \frac{v_i^2}{gR} \right). \quad (11)$$

Additional dayside ionization will result if the solar wind particles interact directly with and are absorbed by the atmosphere (Luhmann *et al.*, 1987). In this case, the solar wind ions will undergo charge transfer to the absorbing neutral particles (mainly O and CO₂). The energetic neutral particles so formed can again become ionized via charge transfer, and so on, until they are eventually absorbed in the atmosphere. However, Kar and Mahajan (1987), using PVO magnetometer data, have shown that there is a strong linear correlation between ionospheric magnetization and solar wind dynamic pressure. That is, there appears to be a linear relationship between the solar wind pressure and the induced magnetic field strength. Cloutier *et al.* (1987) have arrived at similar conclusions by constructing a model of ionospheric motions on the dayside; the model was

based on the integration of the plasma flow equations. They showed that the steady-state solar wind interaction requires a downward flow of the ionospheric plasma together with the downward transport of the interplanetary magnetic field into the ionospheric region. Hence, one can expect the dayside ionosphere to be magnetized, the intensity depending upon the solar wind dynamic pressure. The result is a considerable degree of screening of the ionosphere from direct interaction with the solar wind particles. Shinagawa and coworkers (Shinagawa *et al.*, 1987; and Shinagawa and Cravens, 1988) have developed one-dimensional models of the magnetization by solving the relevant Maxwell equations. However, their work is outside the scope of the present paper (see Luhman and Cravens, 1991).

Some studies by Whitten and coworkers (Whitten *et al.*, 1984; McCormick *et al.*, 1987) have employed models which included vertical flow. They yielded decreasing velocities with increasing SZA on the nightside because of the inclusion of the vertical component. More detailed studies are in progress and will be reported elsewhere.

Suprathermal electron streams have been observed on the nightside by Gringauz *et al.* (1979, 1983) and by Knudsen and coworkers (Spenner *et al.*, 1981; Knudsen and Miller, 1985). The source of these streams is unclear although they are undoubtedly partly of solar wind origin. They are sufficient to produce enough ionization to make an observable ionosphere when ion transport from the dayside is shut off by a low ionopause at the terminator (Knudsen *et al.*, 1987; Knudsen, 1988).

2.2. HEATING

Ion motions are, of course, driven by thermal energy sources. These include photoelectron heating, Joule heating, solar wind-induced heating via wave interactions, precipitating electrons (on the nightside) chemical heating of the ions via the exothermic reactions of CO_2^+ and O^+ (Rohrbaugh *et al.*, 1979), and elastic collisions with the hot oxygen component (Gombosi *et al.*, 1980; Knudsen, 1989).

All of them undoubtedly contribute to electron and ion heating in varying degrees which are still unclear. McCormick *et al.* (1976) computed the photoelectron heating rate by solving a one-dimensional form of the Boltzmann equation. The resulting electron temperatures at high altitude were about 2000 K, roughly one-half to two-thirds that observed by the Pioneer Venus spacecraft (e.g., Miller *et al.*, 1980). If one assumes that the ion gas is heated only by collisions with thermal electrons, the computed ion temperatures are less than 2000 K, again in disagreement with observations (e.g., Miller *et al.*, 1980). Obviously, additional energy sources must be present. This conclusion is reinforced by the observed high temperatures of the nocturnal ion and electron gases, which require the presence of heat sources (Whitten *et al.*, 1986; Singhal and Whitten, 1986). In this context, chemical heating is undoubtedly small and influences the thermal structure of the ion gas only at rather low altitudes (~ 200 to 250 km on the dayside). The relative contribution of Joule heating, precipitating electrons, and heating via wave interactions is not known. The problem of the missing ion heat source on the dayside may have been solved in part by some recent work by Knudsen (1989). He evaluated the heat supply by collisions of O^+ with hot oxygen atoms (Nagy *et al.*, 1981; Nagy and

Cravens, 1988) and obtained a thermal flux supply of $1 \times 10^8 \text{ eV cm}^{-2} \text{ s}^{-1}$ which he claims to be sufficient to account for the observed dayside ion temperature. The hot oxygen component appears to be an insufficient heat source for the nightside ion gas, however. Because the topic of ion energetics is properly the subject of a separate review (Brace and Kliore, 1991), it is not discussed further here.

2.3. IONIZATION

The principal source of ionization for any planetary atmosphere is solar EUV radiation, although streams of ionizing particles of solar origin may also contribute in a major way. The solar ultraviolet radiation can itself vary by a factor of four over the solar cycle, thus changing the upper ionospheric ion density by a similar ratio since the major neutral species there is atomic oxygen. These variations have a pronounced influence on the nightside ionosphere especially, because they control the height of the ionopause and thus the size of the annulus through which the plasma can flow from dayside to nightside (Knudsen *et al.*, 1987; see Section 5.3).

Numerous observations of the solar EUV spectrum have been made in recent years (e.g., Heroux and Hinteregger, 1978; Hinteregger, 1979, 1981, are representative). As mentioned previously, Oppenheimer *et al.* (1981) have also inferred EUV variations over solar cycle 21 from observations of He^+ in the Earth's upper ionosphere. The principal result of their work is the determination that the variation of EUV irradiance from solar minimum to maximum was by a factor of about four.

2.4. NEUTRAL ATMOSPHERE

The structure of the neutral atmosphere is very important to ionospheric dynamics because of the drag exerted on horizontally moving ions and the strong influence on vertical diffusion. As a result of numerous observations made by the Pioneer Venus Orbiter (PVO) (Keating *et al.*, 1980; Niemann *et al.*, 1980) knowledge of the structure of the upper atmosphere of Venus during periods of solar maximum is quite complete. These observations have been supplemented by observations of the Pioneer Venus entry bus (Von Zahn *et al.*, 1980) during morning conditions near solar maximum. Composition measurements of Niemann *et al.* (1980) and of Von Zahn *et al.* (1980) were made with the aid of neutral ion mass spectrometers, while the atmospheric density measurements of Keating and coworkers were obtained from observed orbital changes caused by collisions with thermospheric particles. Unfortunately, there exists no reliable data for solar minimum conditions.

Hedin *et al.* (1983) have constructed a detailed model of the thermosphere (corresponding to solar maximum) based on neutral mass spectrometer observations of Niemann *et al.* (1980). The model was constructed with the aid of a spherical harmonic expansion; since the series was truncated at the fifth order, the rapid changes near the terminator are not faithfully represented. Figure 2 shows concentration profiles of the principal constituents at noon and at midnight as predicted by the Hedin model. Figure 3 portrays the approximate variation of the exospheric temperature with solar zenith angle; the 'ripples' in the curve are an artifact of the truncation of the series

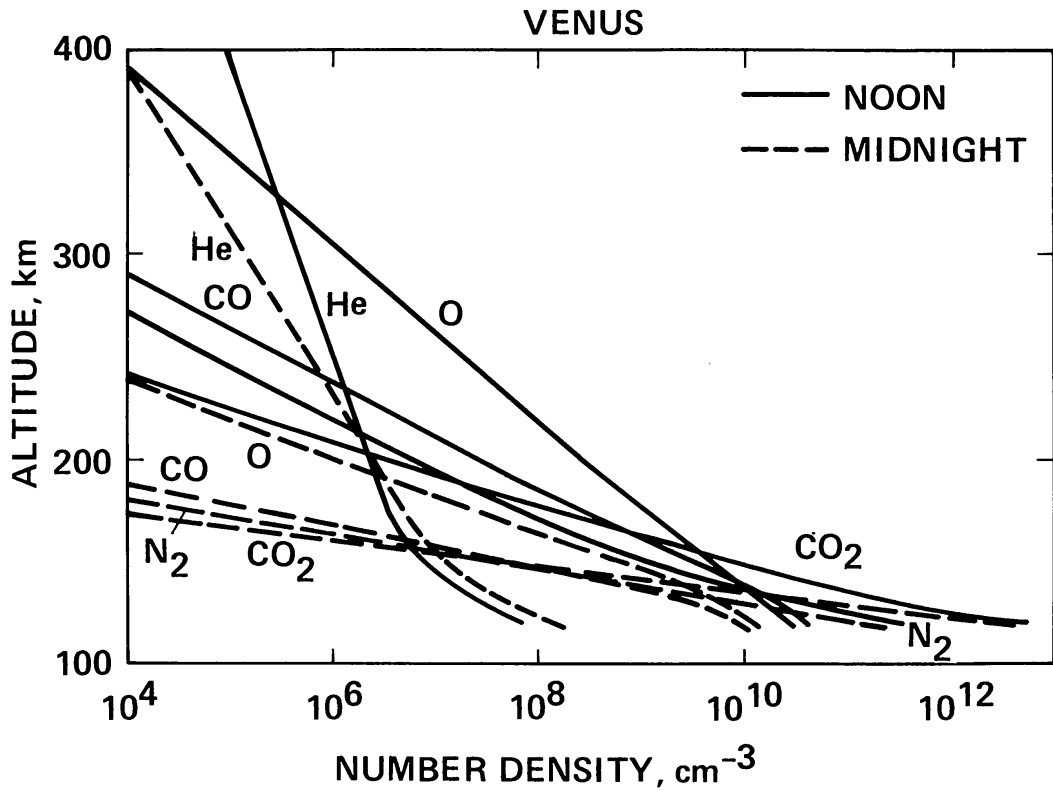


Fig. 2. The day- and nightside neutral constituent number densities on Venus as predicted by the model of Hedin *et al.* (1983).

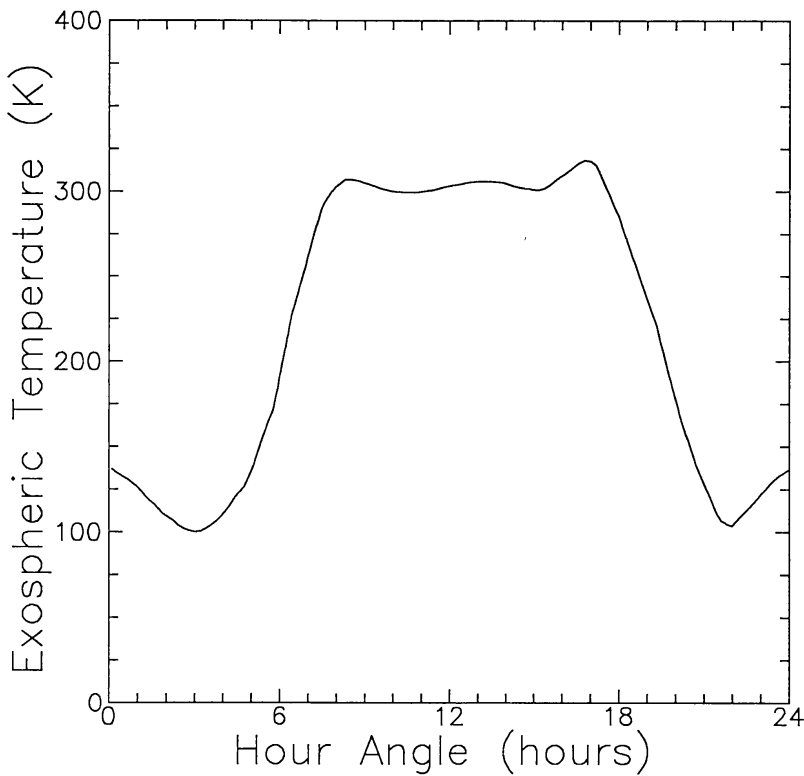


Fig. 3. The Venus exospheric temperature as a function of solar zenith angle (after Hedin *et al.*, 1983).

expansion. McCormick *et al.* (1987) have developed a slightly different model atmosphere based on ionospheric observations coupled with a two-dimensional ionospheric model. However, their results are somewhat ambiguous because the presence of a magnetic field is expected to affect ion diffusion in a manner similar to that of an increased atmospheric density. The Keating model (Keating *et al.*, 1980) based on orbital observations and dynamics differs from that of Hedin and coworkers in that the thermospheric temperatures at the terminator that are predicted by the former are close to the mean nighttime values. Since there is no good theoretical basis for such variation, it appears that the Hedin *et al.* (1983) thermospheric diurnal temperature variation is to be preferred.

Nagy and coworkers (Nagy *et al.*, 1981; Nagy and Cravens, 1988) have suggested that energetic oxygen atoms are present in the high atmosphere of Venus as a result of the dissociative recombination of O_2^+ . Their early model calculations have proved to be about an order of magnitude too large, but their more recent values (Nagy and Cravens, 1988) ($\sim 10^3 \text{ cm}^{-3}$) are very close to the values deduced from Pioneer Venus Orbiter EUV observations (Paxton and Stewart, 1988). As Knudsen (1989) has shown, the smaller values are still large enough to substantially influence the dayside ionospheric thermal balance.

3. Observations

3.1. METHOD OF ION VELOCITY MEASUREMENTS

The method by which the Pioneer Venus ORPA measures the velocities of the ions is described by Knudsen *et al.* (1979, 1980c). Basically, it is the measurement of the mean O^+ kinetic energy in three directions on three successive spacecraft rotations. The mean energy is determined by a least-squares fit of a travelling Maxwellian distribution function to the measured current/voltage curve.

The ORPA is mounted at an angle of 25° from the spacecraft spin axis. In its most common mode, it is programmed to record data when the instrument is in the spacecraft ram direction in one spin period, 45° earlier in the next spin period, and 45° later in the third spin period. The line-of-sight of the second and third measurements are thus separated from the ram measurement line-of-sight by about 18.6° . The resolution of these three measurements into a Cartesian coordinate system provides the vector sum of the spacecraft and ion velocities. After subtracting the velocity of the spacecraft, the resultant vector formed from the three line-of-sight velocity measurements corresponds to the ion velocity of the ambient ions.

The spacecraft speed of approximately 9 km s^{-1} at periapsis insures that the flow is directed into the ORPA for measurements when the spacecraft is in the ionosphere. Since the spacecraft spin period is about 12 s, this orbital velocity also means that the spacecraft has traveled about 100 km of horizontal distance between measurements. In spite of approximately 200 km of horizontal distance separating the first and last measurement, the vectors formed from three successive measurements are usually

self-consistent and in a nightward direction. The results suggest that there is a primary flow of ions that is global in scale, in spite of significant small-scale spatial and temporal variations.

The other instrument on board the Pioneer Venus Orbiter that is sensitive to ion velocity is the ion mass spectrometer (OIMS) (Taylor *et al.*, 1980a). The instrument is a Bennett RF spectrometer, and performs a mass analysis by an initial RF voltage that increases the kinetic energy of resonant ions, followed by a retarding voltage that rejects all ions that have not achieved the maximum energy. A servo system is used to tune the instrument for changes in ion-ram velocity and spacecraft potential. Monitoring this servo loop gives a measure of the ram energy of the ions, from which the component of ion drift in the ram direction can be derived.

In addition to the measurement of ionospheric ram velocities by the OIMS, Taylor *et al.* (1979, 1980b, 1981) have also reported the detection of superthermal ions at the top of the ionosphere. The OIMS cannot discriminate between ions of different masses at these velocities, but rather infers the presence of superthermal ions by detecting an ion component with kinetic energies that are too high for the OIMS to measure.

3.2. MEAN ION VELOCITIES

This section is a statistical summary of the ion velocity measurements. A coordinate system is used to present the majority of the data in which all positions and velocities have been projected onto a plane containing the Sun, Venus, and the spacecraft. The two coordinates of this projection are solar zenith angle (SZA) and altitude. This is a useful representation since most ionospheric quantities are nearly axially-symmetric about the Sun–Venus axis.

The method of presentation of ion velocities for use in studies of the Venus ionosphere can itself influence the conclusions drawn from the data. Presenting data in coordinates of SZA and altitude is equivalent to assuming axial symmetry about the Sun–Venus axis. Although this is apparently the predominant symmetry in the Venus ionosphere, non-negligible asymmetries exist. The most apparent is between dawn and dusk hemispheres.

The ionopause and any phenomena that are strongly affected by the solar wind would, because of the motion of the planet along its orbit, tend to have an axis of symmetry that is aberrated by about 5° toward the dusk terminator. But, since most of the processes that affect, or are affected by, ion dynamics occur below the ionopause, the different incident directions of the solar electromagnetic radiation and the solar wind are not considered in the discussion.

The ion velocity, like most of the quantities that have been measured in the Venus ionosphere, is variable on a time-scale shorter than the orbital period of the spacecraft. Figure 4 shows ion velocities measured on orbits 448 and 449, separated by only about 24 hours. Periapsis of the two orbits were at solar zenith angles of 61° and 62° , respectively. Both show nearly constant nightward flow to an altitude of about 400 km. Above 400 km, velocities from orbit 449 deviate from this nightward velocity in a manner that is probably not realistic. The indication is that there are major changes in the direction of flow within the 200-km interval of the velocity measurement. The

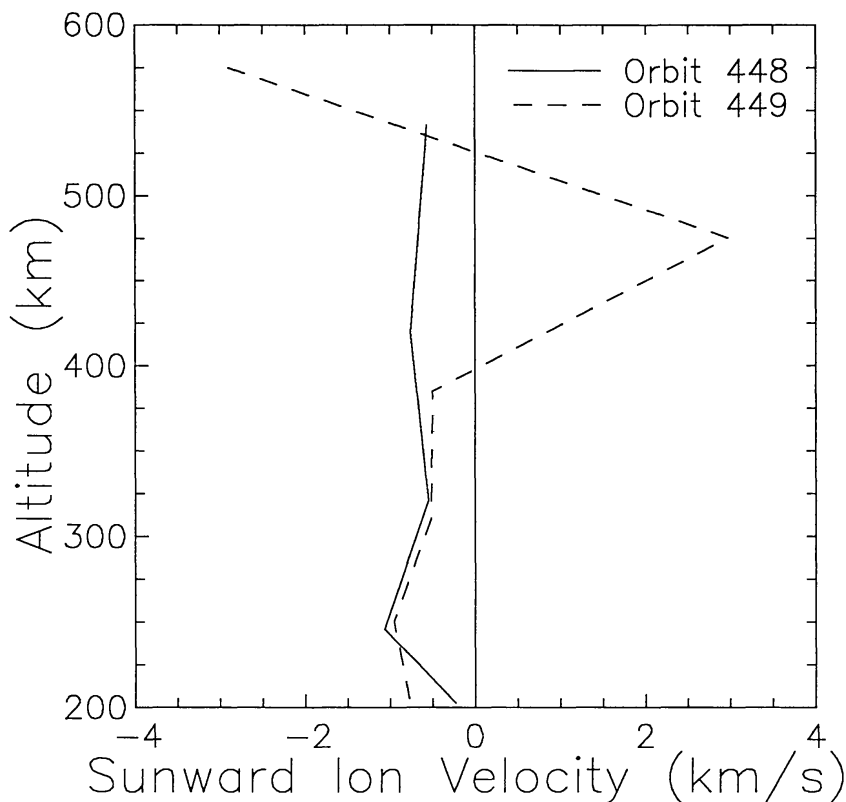


Fig. 4. Horizontal component of O^+ velocities in orbits 448 and 449, possibly showing the effect of changes or shears in the ion velocity on a scale that is shorter than that required for making a vector measurement by the ORPA. SZA at periapsis is approximately 62° .

velocities in orbit 448, on the other hand, progress smoothly from the periapsis to ionopause.

Figure 5 shows average ion velocities measured in the outbound leg of the Pioneer-Venus orbit during the first 3.5 years of the Pioneer-Venus mission (Knudsen *et al.*, 1982a). The ionization is flowing generally in an anti-Sunward direction with average speeds that approach 4 km s^{-1} near the terminator. Although significant variability is observed in non-averaged measurements from individual orbits, the average ion velocity varies smoothly at solar zenith angles less than 140° .

Measured line-of-sight velocity components can be quite large on the nightside, and the resolved velocity vectors are not as consistently anti-Sunward as at smaller SZA. Averaging such large, but randomly directed vectors results in a vector field with small magnitudes. Such is the case with ion velocities within 40° of the anti-solar point which appear to be small in Figure 5, but, because of more chaotic conditions which prevail there, can actually represent averages of large velocities. The large velocities are themselves probably in error, since any velocity shear occurring within the 200-km interval of the three measurements would invalidate the result.

The velocity vectors shown in Figure 5 are from the outbound legs of the Pioneer Venus orbit. The ion velocity measured by the ORPA has a consistent downward component on the inbound leg of the orbit, and is nearly horizontal on the outbound

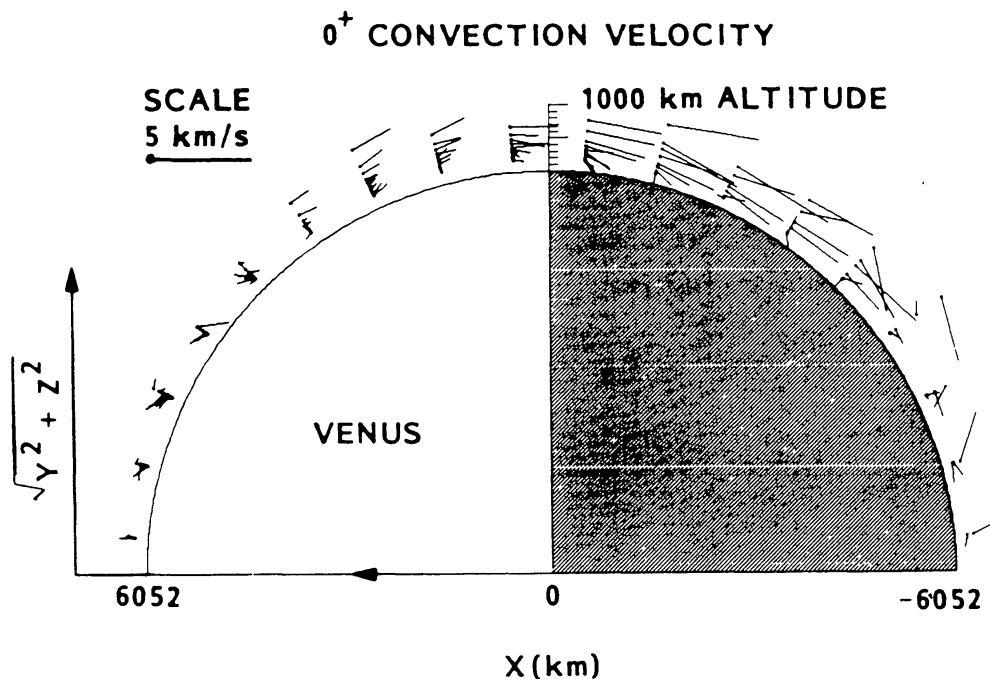


Fig. 5. Average ion velocities measured in the outbound leg of the orbit during the first 3.5 years of the Pioneer Venus mission (after Knudsen *et al.*, 1982a).

leg. Some downward flux is expected on the basis of theory (Cravens *et al.*, 1984; Cloutier *et al.*, 1987), but the measured velocities are much larger than expected. Although no systematic error has been determined from the ORPA data, apparent large vertical velocities may well be in error as a result of the large angles of attack of the instrument on the inbound leg. The angles of attack are smaller and are within acceptable limits on the outbound leg. For this reason, data from the outbound leg only are used for this statistical presentation. The use of outbound data has the additional advantage that it is from the vicinity of the equator, which reduces the ambiguity when asymmetries in the flow are isolated.

Averages of the horizontal anti-Sunward component of the O^+ velocity field are shown in Figure 6, together with theoretical results. These values are averages of all the outbound data measured by the Pioneer-Venus ORPA. The data have been smoothed to remove random variations which are probably related to velocity shears. The amount of smoothing used for the velocity measurements shown in Figure 6 are represented by 'error bars' shown at selected data points. The error bars indicate the range within which two-thirds of the measurements fall. This is representative of the measurements on the dayside, and to about 140° SZA on the nightside. Nightward of 140° , the velocity vectors become randomly oriented and cannot be accurately represented by average vectors.

In addition to the axially-symmetric bulk flow of the ionosphere, the OIMS (Taylor *et al.*, 1980b) measured regions of superthermal ion flow. The superthermal flow region begins, in most cases, in the region of the inopause and extends to higher altitudes. Figure 7 shows an example from the outbound leg of an orbit on the nightside of Venus.

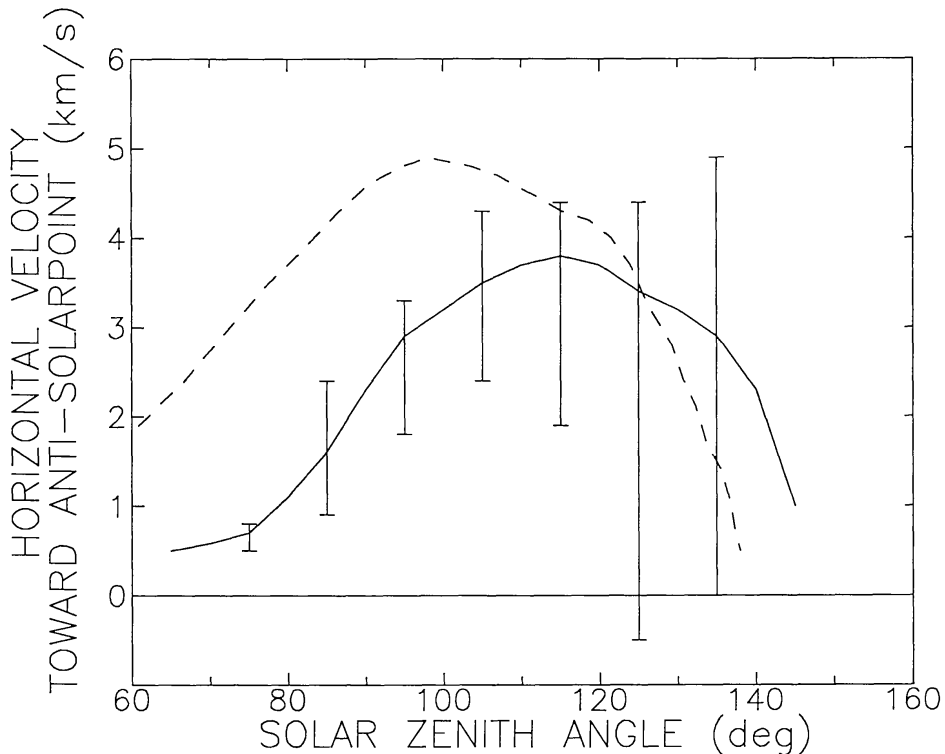


Fig. 6. Observed and calculated horizontal ion velocities toward the anti-solar point at 400 km altitude. The dashed curve was obtained with the simplified model of Singhal and Whitten (1987). The observed data (obtained from the ORPA experiment on PVO) are the average values, while the error bars show the standard deviation from the mean.

V_f is the component of ion velocity parallel to the spacecraft axis and directed into the instrument, as measured by the OIMS. Superthermal ion flow, marked by the shading, is observed mainly above the ionopause, but is also seen in isolated regions at lower altitudes. Taylor *et al.* (1980b) also reported detached regions of superthermal ionospheric ions above the ionopause, presumably associated with patches of detached ionization or streamers.

3.3. MAGNETIC FIELD EFFECTS

While Venus has quite convincingly been shown to possess no intrinsic magnetic field (e.g., Russell *et al.*, 1980), induced magnetic fields certainly are present within the ionosphere. In fact, the field intensities sometimes exceed 100 nT (Luhmann *et al.*, 1987). Such fields must influence plasma motions because the ratio of thermal plasma pressure to magnetic pressure (usually referred to as 'beta') may be substantially less than unity. In fact, plasma motions and the magnetic field are closely coupled and should be treated as a single system, at least near the subsolar point (Cloutier *et al.*, 1987; Shinagawa *et al.*, 1987; Shinagawa and Cravens, 1988). A review of the characteristics of the magnetic field and its effects on the Venus ionosphere is presented elsewhere (Luhmann and Cravens, 1991). Under quiescent solar wind conditions the ionosphere of Venus is nearly field-free and the modeling techniques described in this paper are thoroughly valid. Under disturbed conditions, however, the plasma motions

PIONEER VENUS OIMS

ORBIT 57

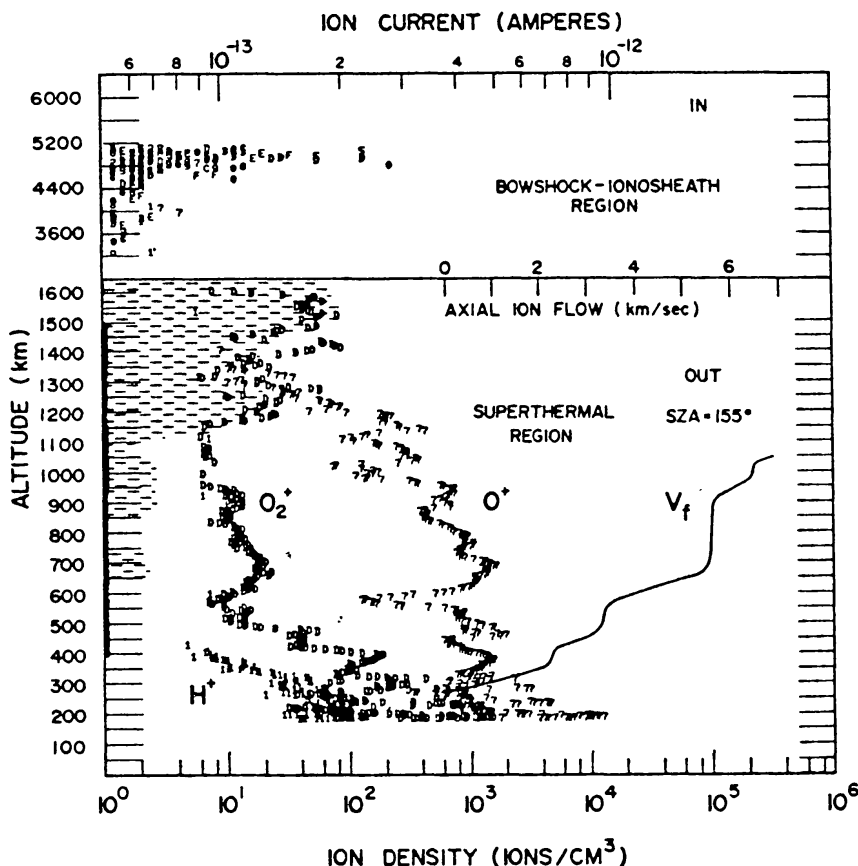


Fig. 7. Measurements by the OIMS during an orbit on the nightside of Venus. The energetic ion currents in the ionosheath have energies greater than 100 eV. Regions of superthermal ion flow (energy = 30–80 eV) are shown by shading. The width of the shading indicates an uncalibrated relative intensity. V_f is the axial component of the ion flow (after Taylor *et al.*, 1980b).

are substantially altered in the subsolar region (e.g., Shinagawa *et al.*, 1987) and two-dimensional modeling techniques are less applicable. As one approaches the terminator, the induced fields tend to drape themselves around the planet, allowing free horizontal flow, but inhibiting the vertical transport of the plasma. This influence was pointed out by McCormick *et al.* (1987) in trying to reconcile the Pioneer Venus retarding potential analyzer measurements of ion density with the neutral atmospheric structure obtained from the neutral mass spectrometer experiment.

4. Theoretical Results

4.1. ION VELOCITIES AND DENSITIES

The trans-terminator plasma pressure gradient of the Venus ionosphere may be sufficient to accelerate the plasma to the observed supersonic speeds (Knudsen *et al.*, 1981; Spenser *et al.*, 1981; Whitten *et al.*, 1982, 1984; Elphic *et al.*, 1984b; McCormick

et al., 1987; Singhal and Whitten, 1987). Crude two-dimensional simulations of the dynamics were first reported by Chen (1977) and by Chen and Nagy (1978), who extended a one-dimensional model to examine horizontal plasma transport as far as the terminator (SZA = 90°). Whitten *et al.* (1982) used an appropriately modified one-dimensional model to look behind the terminator to SZA = 102.5° . Both sets of investigators found theoretical evidence for very substantial transport of ions from the dayside to the nightside hemisphere as a result of the transterminator plasma pressure gradient.

The next step was the development of genuine two-dimensional models which would predict ion densities far into the nightside hemisphere. Cravens *et al.* (1983) constructed a model which predicted the densities of ion species (O^+ , O_2^+ , CO_2^+ , H^+ , C^+ , He^+ , NO^+) up to SZA = 160° , using a rough empirical fit to the ion velocity field observed by Knudsen *et al.* (1981). The fit was designed such that the largest velocities occurred near the equator and then decreased sinusoidally for larger SZA; the simulated velocity profiles increased linearly with increasing altitude. Although a major advance in predicting nocturnal ion densities (the computed number densities were in very good agreement with observed values), the Cravens model was not self-consistent in the sense that it did not provide for feedback of the computed ion densities into the velocity field.

Theis *et al.* (1984) and Elphic *et al.* (1984b) took a different tack, constructing empirical models of the plasma flow dynamics using measured electron densities and temperatures from the electron temperature probe (OETP) on board Pioneer Venus Orbiter (PVO). Their models consisted of simplified (one-dimensional) solutions to Equation (1); ion temperatures needed to compute sonic velocities were assumed to be equal to the means of the electron and neutral atmosphere temperatures. They also included viscosity in their calculations of the horizontal velocity fields. Their results, shown in Figure 8, indicate the occurrence of high velocities at high altitude near the

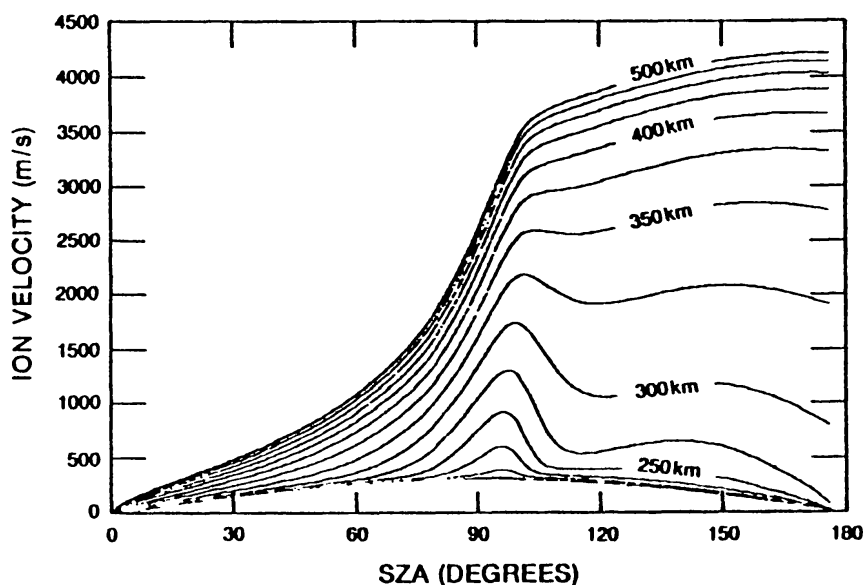


Fig. 8. Solution to the momentum equation, neglecting plasma viscosity effects, but including ion neutral drag (after Theis *et al.*, 1984).

anti-solar point, an impossible condition. This behavior led them to conclude that viscosity, due either to ion collisions, magnetic fields or both, is essential to slow the plasma flow on the nightside.

In a later modification to those models, Elphic *et al.* (1984b) concluded that viscosity had not been properly included in the calculations. As a result, they concluded that viscosity is not very important to the ion dynamics and suggested that downward advection on the nightside is the dominant process for slowing the plasma. They also suggested the possibility of the occurrence of turbulence as a result of the large velocity shears. Some recent unpublished work by Whitten and Singhal supports this conjecture. Figure 9 shows vertical profiles of the horizontal ion velocity calculated by Elphic *et al.* (1984b). They agree reasonably well with the velocity profiles calculated by Whitten *et al.* (1984) using a different approach.

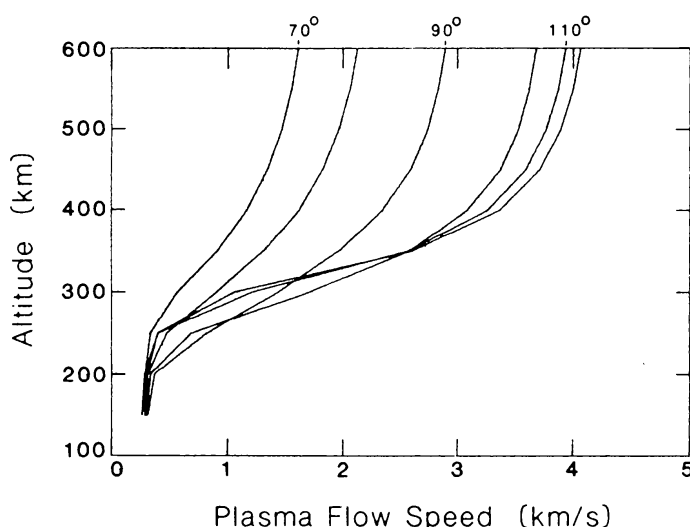


Fig. 9. Flow speeds for the inviscid solution versus altitude for six solar zenith angle locations near the terminator. Note the strong velocity shear that develops at 300 km altitude just beyond about 90° (after Elphic *et al.*, 1984b).

Whitten and coworkers (Whitten *et al.*, 1984; McCormick *et al.*, 1987) developed a two-dimensional model in which ion number densities (O^+ , O_2^+ , CO_2^+ , CO^+ , He^+) and velocities (both horizontal and vertical) were computed. The horizontal velocity component and ion number densities are shown in Figures 6 and 10, respectively. It is apparent from the solid curve in Figure 6 that the computed horizontal component of the ion velocity is in very good agreement with observations; admittedly, the error bars on the nightside data are quite large, accommodating a broad range of allowed velocity predictions.

In the course of varying the parameters which influence the ion density, it was discovered that the computed nocturnal ion densities are very sensitive to the structure of the neutral atmosphere and to the presence of magnetic fields (McCormick *et al.*, 1987). In addition, the ion velocities are sensitive to the ratio between dayside and

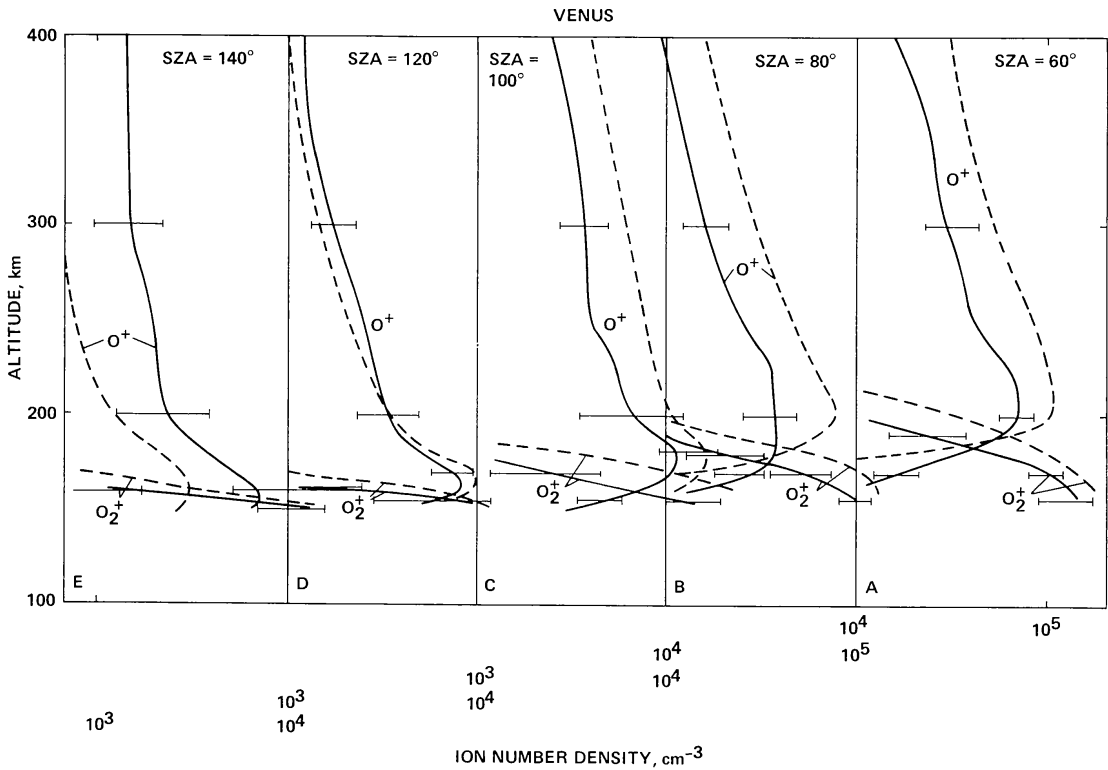


Fig. 10. Observed (solid lines) and calculated (broken lines) O^+ and O_2^+ densities at various solar zenith angles. The observed data (obtained from the ORPA experiment on board Pioneer Venus orbiter) are the median (50% quartile) values while the error bars are illustrative of the 25 and 75% quartiles.

nightside major ion number densities at high altitude. We obtain from Equation (9) (approximately)

$$v^2 = c^2 \ln [N_d/N_n], \quad (12)$$

where $c = [2kT_p/m]^{1/2}$ is the plasma sonic velocity, and N_d and N_n are the dayside and nightside high altitude major ion number densities, respectively.

As an alternative to the pressure gradient-induced plasma flow, Perez-de-Tejada (1982) suggested that the plasma is dragged along by the solar wind which is coupled to the ionosphere through viscous forces. Unfortunately, for Venus it is not possible to distinguish between flow induced by a pressure gradient and flow induced by the solar wind interaction. It may be possible to do so for Mars (Singhal and Whitten, 1988), which may in turn shed more light on the corresponding situation with Venus. Some unpublished calculations have indicated that the shapes of the vertical profiles of velocity are not very different for the two mechanisms, certainly not within the measurement uncertainties.

4.2. TRANS-SONIC FLOW

It is useful to describe here the mechanism for the transition from subsonic to supersonic flow. In order that the flow become supersonic, it is essential that a converging-diverging nozzle be simulated in some way (e.g., Banks and Kockarts, 1973). In the Venus

ionosphere the 'convergence' of the nozzle is supplied by the steady addition of mass (inertial drag) to the flowing ion gas by photoionization of atomic oxygen and by resonant charge exchange of O^+ with the hot atomic oxygen component at high altitude. The divergence on the nightside results from mass removal by vertical transport and ion chemical reactions low in the ionosphere. The nozzle 'throat' is located at or just behind the terminator. Solution of the steady-state momentum equation (1) requires special techniques to integrate through the 'critical points' at which the plasma velocity is equal to the sonic velocity. However, a time-dependent approach like that used by Whitten *et al.* (1984) avoids this problem by allowing the solutions to 'relax' to the steady-state.

4.3. VELOCITY CONVERGENCE ON THE NIGHTSIDE

As discussed previously, Singhal and Whitten (1987) constructed a spectral model of the dynamics of the Venus ionosphere in which they reduced the ion momentum equation to one dimension. Attenuation of the plasma velocity on the nightside was effected by multiplying the computed velocity by a factor $\sin \Theta$. This was in contrast to the suggestion of Theis *et al.* (1984) that ion viscosity is necessary to account for the slowing of the plasma on the nightside. In their calculations, Singhal and Whitten (1987) investigated the effects of viscosity on plasma velocities and found that for reasonable values, the velocity would be reduced by about 10% or less. The plasma velocities obtained by Singhal and Whitten without introducing viscosity are represented by the broken curve in Figure 6. Since it converges toward the anti-solar point, the velocity of the plasma must eventually vanish unless the electron-ion gas is convected out through the wake and lost to the planet. As the plasma velocity decreases through the transonic range, a shock wave should form. There is some evidence for shock wave formation at an SZA of about 140° ; specifically, the sudden fluctuations observed in the velocity field. Knudsen *et al.* (1980a) suggested that the rapidly increasing ion temperatures occurring in the same region indicates that the cessation of mean nightward flow is due to a recompression shock in the plasma. However, Whitten *et al.* (1986) showed that the same amount of kinetic energy would be converted into heat from an adiabatic compression as from a recompression shock. Neither the abrupt decrease in velocity on the nightside nor the high ion temperatures at the anti-solar point have been successfully modeled.

4.4. PLASMA TEMPERATURE

Calculations of dayside temperatures of the ion and electron gases on Venus were done initially by Whitten (1969), who used a very simple model of electron heating efficiency and various assumptions about the heat flux from the solar wind. McCormick *et al.* (1976) used greatly improved calculations of the electron heating efficiency to arrive at high altitude dayside electron temperatures (4000 K) that proved to be quite close to those observed by Pioneer Venus Orbiter (PVO). Chen (1977) and Chen and Nagy (1978) extended the one-dimensional model computations as far as the terminator, the results of which were also quite close to the values observed by PVO (e.g., Miller *et al.*, 1980).

The next step was the development of two-dimensional models which could simulate heat transport to the nightside by means of the bulk flow of the plasma. Following a study of the energetics of the nightside ionosphere (Knudsen *et al.*, 1980b), Bougher and Cravens (1984) constructed a model which solved the ion energy equation (Equation (2)) by finite differencing in the vertical and horizontal directions. They used the plasma bulk velocity fields employed previously by Cravens *et al.* (1983) to compute ion number densities on both the day and the nightsides. Results obtained with a solar zenith angle-dependent horizontal velocity patterned after the ion velocities observed by Knudsen *et al.* (1981) are shown in Figure 11. It should be noted that Bougher and Cravens (1984) assumed the absence of nocturnal heat sources for the ion and electron gases. As a consequence, their predicted ion temperatures on the nightside (2000 K) were well below the observed temperatures in the central nightside ionosphere of 5000 K or larger (Miller *et al.*, 1980) but consistent with results of Singhal and Whitten (1986) for the case in which no external source was supplied to the nightside ionosphere.

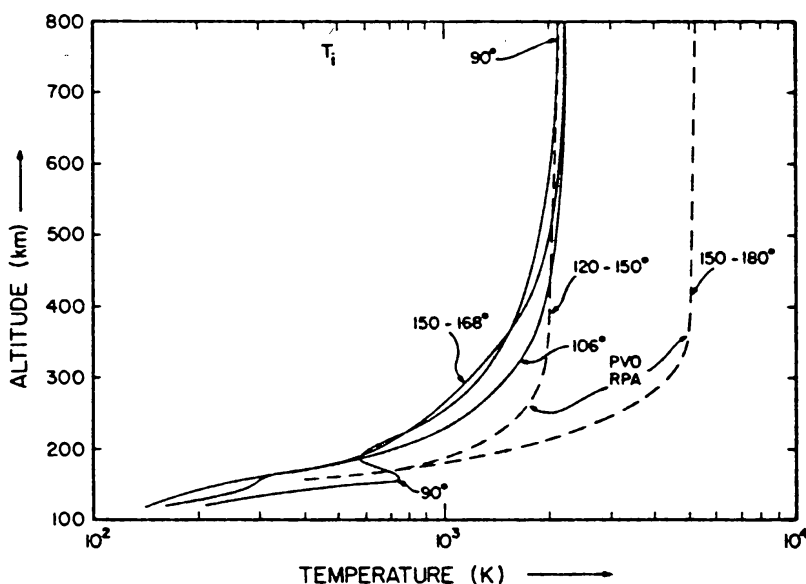


Fig. 11. Calculated (solid lines) and observed (broken lines) ion temperatures for several solar zenith angles. The measured velocities are from the observational data of Knudsen *et al.* (1981) (after Bougher and Cravens, 1984).

Whitten and coworkers (Whitten *et al.*, 1986; Singhal and Whitten, 1986) developed spectral models which solved the ion and electron energy equations. That is, the solar zenith angle-dependence of the various parameters including the temperature were expanded in Legendre polynomials (axial symmetry was assumed), the argument of which was the cosine of the solar zenith angle. Vertical derivatives were simulated by finite differencing and horizontal plasma velocities were simulated by using the predictions of Whitten *et al.* (1984). Like Bougher and Cravens (1984), Whitten and coworkers found that the predicted nighttime temperatures were much too low (1500 K) and that a nighttime heat source of $1-2 \times 10^{-4}$ ergs $\text{cm}^{-2} \text{s}^{-1}$ was necessary to satisfy observa-

1991SSRV...55..165M

tions. They also found that as the plasma flowed toward the nightside, the ion temperature first fell (to 1000 K with no nocturnal heat source) as a result of adiabatic expansion and then rose due to compression. This effect did not appear in the results of Bougher and Cravens (1984). The computed high altitude ion temperatures obtained by Singhal and Whitten are shown in Figure 12.

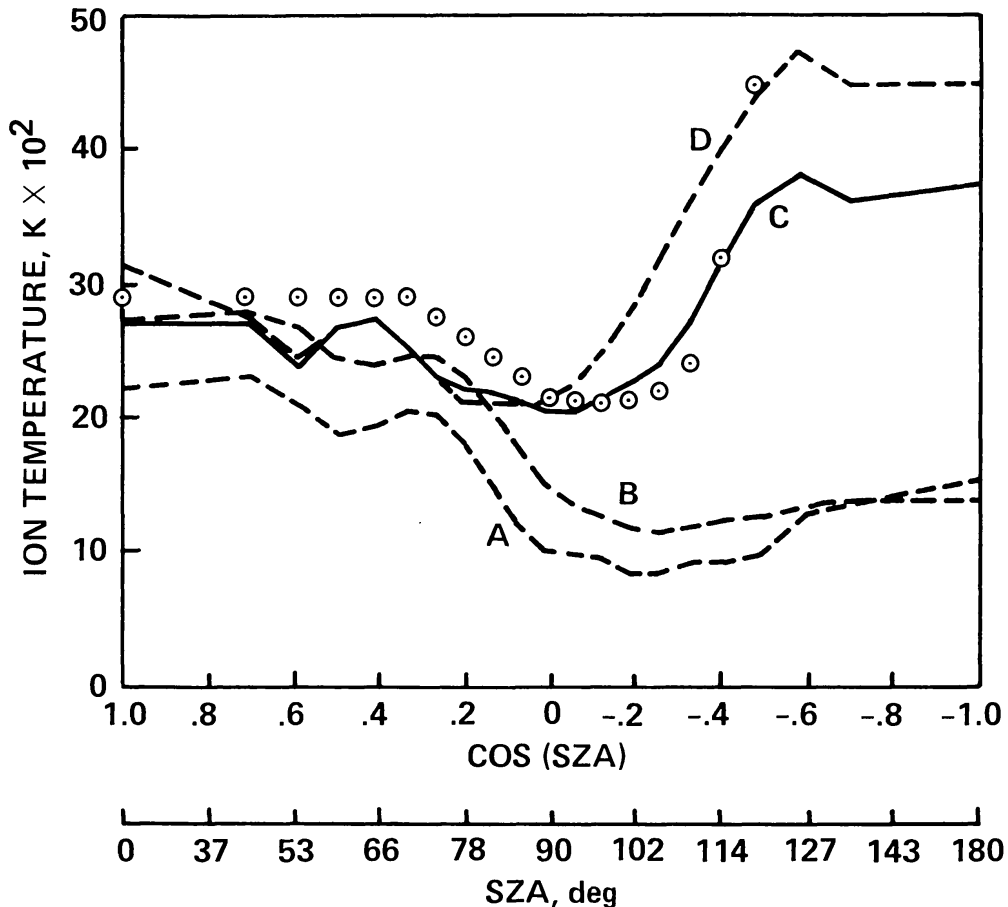


Fig. 12. Computed ion temperatures at high altitudes. Curve *A*: no external heat source to ion gas; curve *B*: heat to ions $1.4 [\cos(\text{SZA})]^{1/4}$ on the dayside, 0 on the nightside; curve *C*: 1.4 uniform on dayside, 1.4/2 uniform on nightside; curve *D*: 1.4 uniform on dayside, 1.4/1.2 uniform on nightside. Circles denote the observed high-altitude ion temperatures. Heating rate units are $\text{eV cm}^{-3} \text{s}^{-1}$ (after Singhal and Whitten, 1986).

5. Temporal and Spatial Variability

5.1. DAWN-DUSK ASYMMETRY

Miller and Knudsen (1987) have presented evidence of a dawn-dusk asymmetry in the ion flow. An example of this asymmetry is the observation that the divergence and convergence points are not coincident with the sub-solar and anti-solar points and are not opposite each other on the day and night sides. Figure 13 shows horizontal O^+ velocities in the ionosphere of Venus projected onto a latitude-longitude grid. The origin

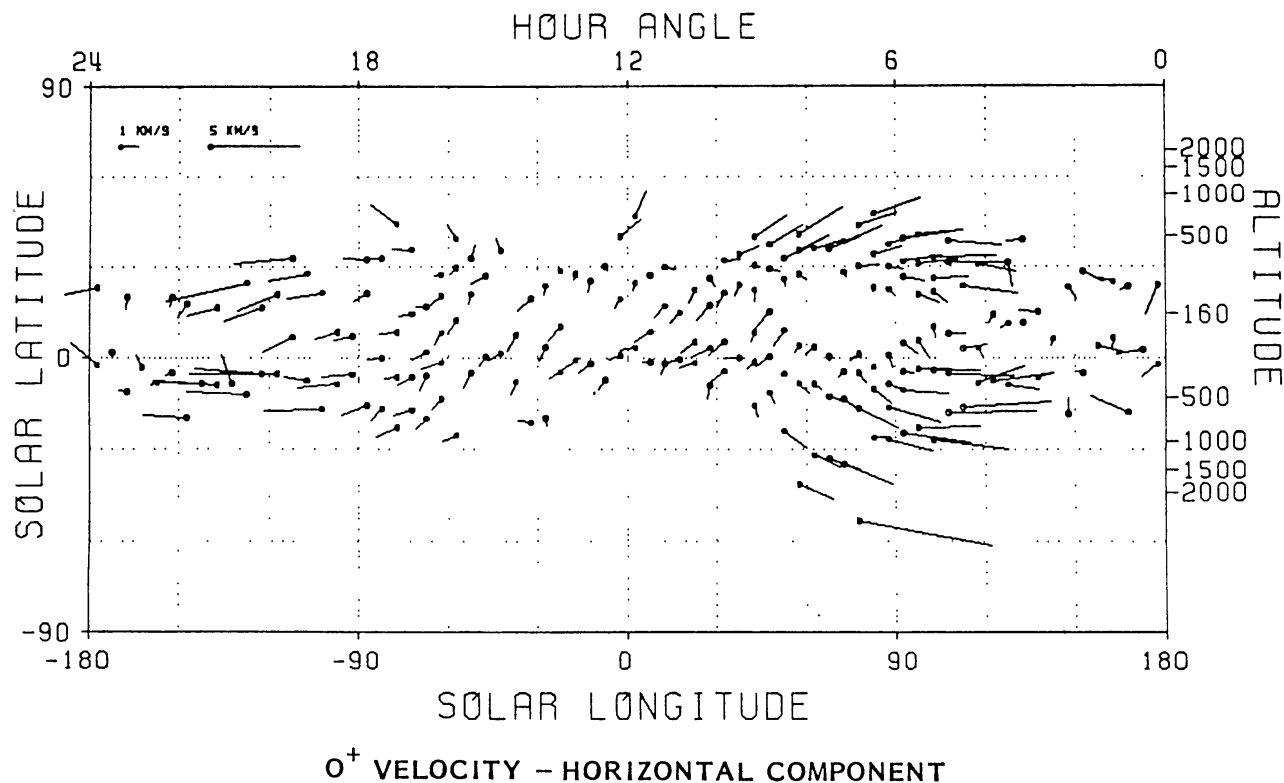


Fig. 13. Horizontal O^+ velocities measured by the ORPA. Periapsis of the PVO orbit is at about 15° N latitude.

is defined as the sub-solar point. Longitude is measured to the east of the sub-solar meridian. Because of the retrograde planetary and atmospheric rotation, the dusk terminator is at -90° and the dawn terminator at 90° . Periapsis occurs at approximately 15° north on this plot, with the inbound leg to the north and the outbound leg to the south. The flow pattern is seen to diverge from the dayside and converge into the nightside. However, the points of symmetry are displaced from the sub-solar and anti-solar points. The center of divergence is east of the sub-solar point, and the center of convergence is west of the anti-solar point. The directions of the displacements of symmetry points is consistent with the addition of a superrotation (or a constant westward velocity) to the axially-symmetric flow.

Figure 14 shows the average zonal component of the horizontal ion velocity at 250 km altitude as a function of the longitudinal angle measured eastward from the sub-solar point. Positive is toward the east. The flow is generally symmetric about the sub-solar meridian, but it can be seen to be offset by a few hundred meters per second toward the west. This dawn/dusk asymmetry can be interpreted as a superrotation of the ionosphere superimposed on the axially-symmetric flow (Miller and Knudsen, 1987). The superrotation is in the same sense as the superrotation of the neutral atmosphere, but is much larger than model calculations of neutral superrotation predict. An explanation of these data based on instrumental or geometric effects that would be interpreted as a 400 m s^{-1} superrotation has been sought, but unsuccessfully.

The neutral atmosphere is observed to superrotate with a zonal wind speed of about

1991SSRV...55..165M

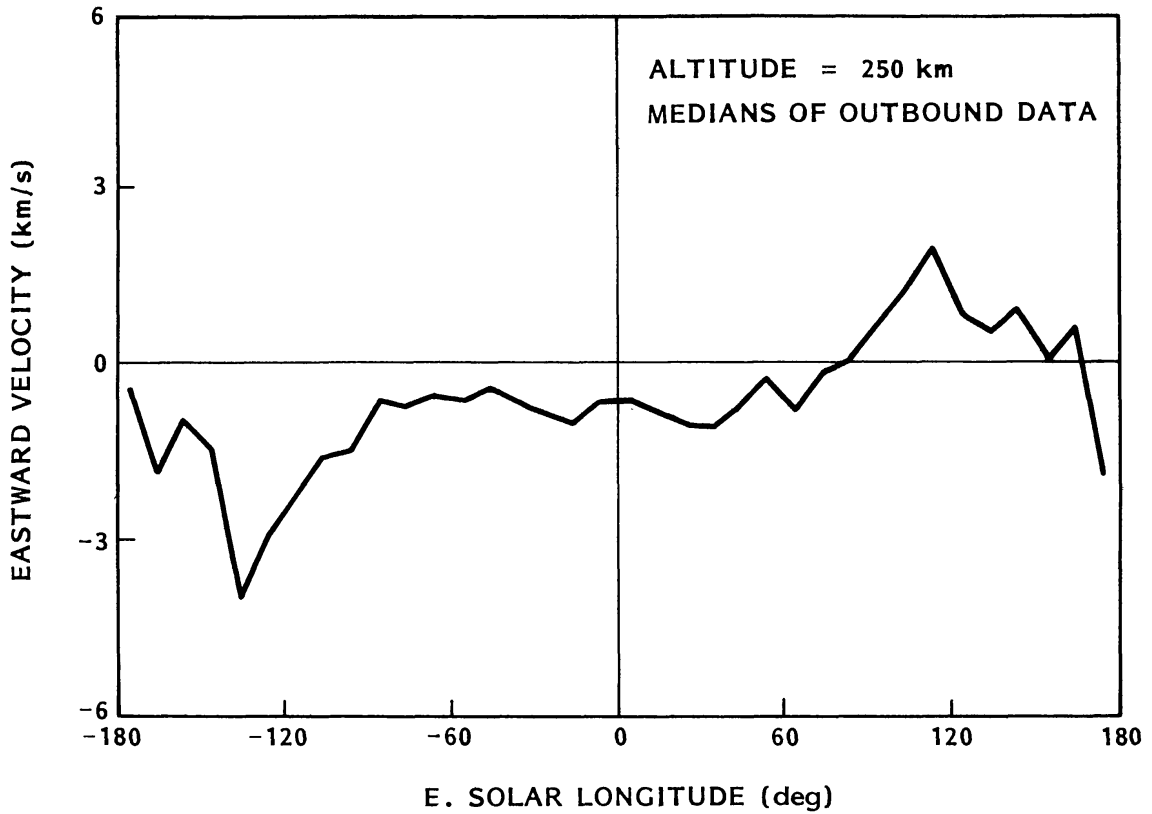


Fig. 14. Average eastward component of the O^+ velocity at 250 km altitude (after Miller and Knudsen, 1987).

100 m s^{-1} or less at the cloud tops (Schubert, 1983). Although this is much less than the apparent superrotation velocity of the ionosphere, it may be that the motion of the neutral atmosphere serves to give a zonal bias to the ionospheric flow and creates a negative pressure gradient from the equator to the poles, resulting in the conditions necessary for the generation of superrotation. The requirements of cyclostrophic balance under such conditions can be estimated, and gives a superrotation velocity of about 1 km s^{-1} (Elphic *et al.*, 1984b).

Figure 15 shows zonal flow of the ions near the equator. These are average velocities determined from the outbound segments of the orbit only. The altitude scale in Figure 15 is expanded by a factor of four to emphasize the flow regions. The darker area is the region where eastward flow is observed; the lighter area, where westward flow is observed. On the dayside, the regions are seen to overlap in the dawn sector, with the boundary between the flow regimes increasing in altitude toward the sub-solar point. On the nightside, the flow appears to undergo a shock that randomized the flow field. The center of the region between the apparent shocks is seen to be displaced toward dawn by about 10° .

The higher density and velocity on the dusk side indicate a greater flux into the nightside. This suggests that a higher ion density might be expected in the post-midnight sector. However, this is not observed. In fact, the only obvious asymmetry in the nightside that can be attributed to superrotation of the ionosphere is the offset of the

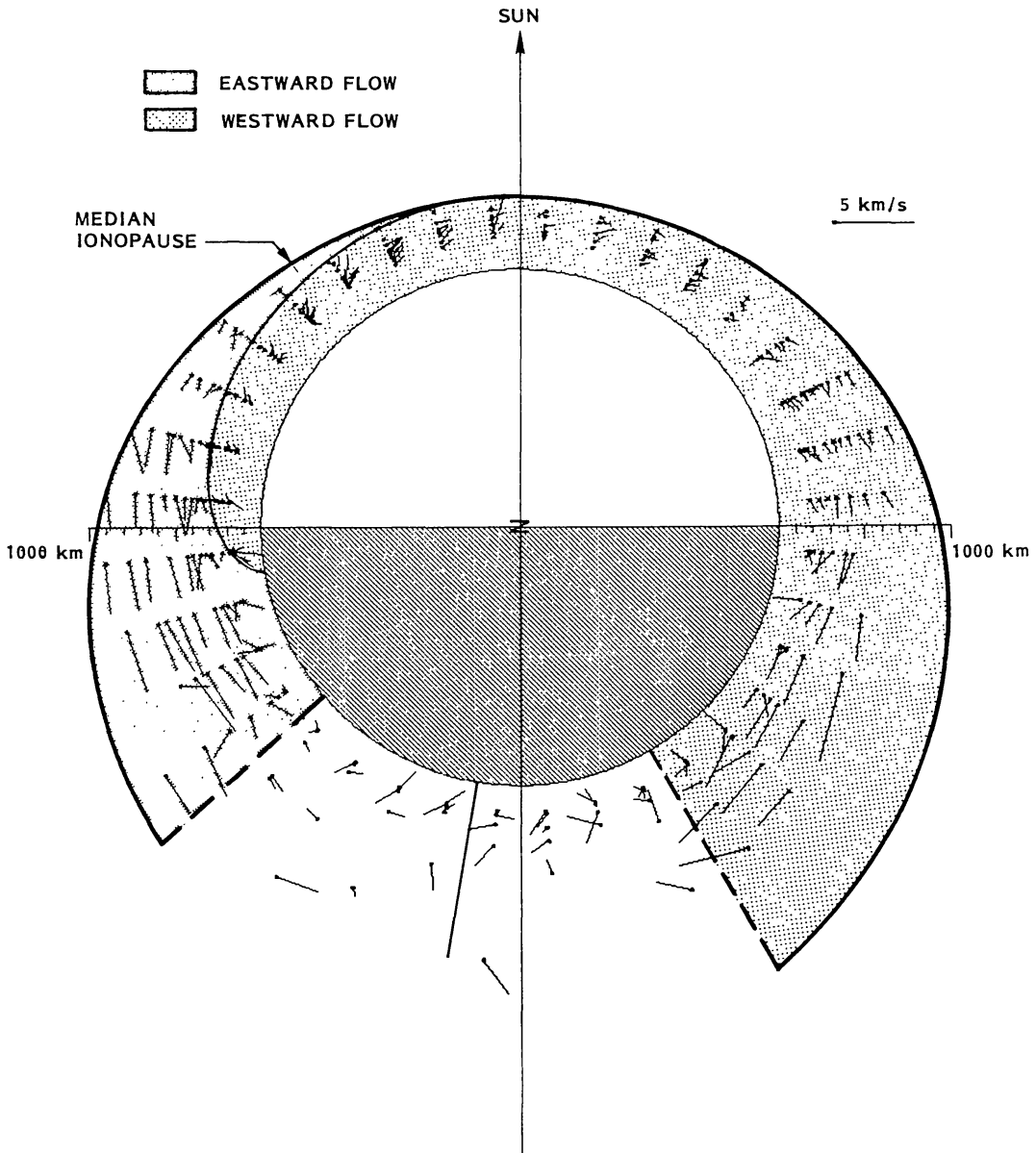


Fig. 15. Anti-Sunward O^+ velocity averages at 10° intervals in longitude. The altitude scale has been exaggerated by a factor of four relative to the planetary radius. The shaded area is approximately defined by the median altitude of the ionopause, with the nightward boundary near the point where the velocities become chaotic. The light and dark shading denote regions where the velocity is predominantly westward and eastward, respectively.

apparent recompression shock, as shown in Figure 15. The other major nightside asymmetry, the relatively high H^+ concentration in the dawn sector, was shown by Mayr *et al.* (1980, 1985) to be caused by changes in the neutral composition resulting from superrotation of the thermosphere.

The measured superrotation component generally decreases with increasing altitude. Figure 16 shows the difference between average nightward velocities compared with calculations based on the momentum equation (Equation (1)). The heavy curve in Figure 16 shows the difference between measured horizontal nightward velocities in the

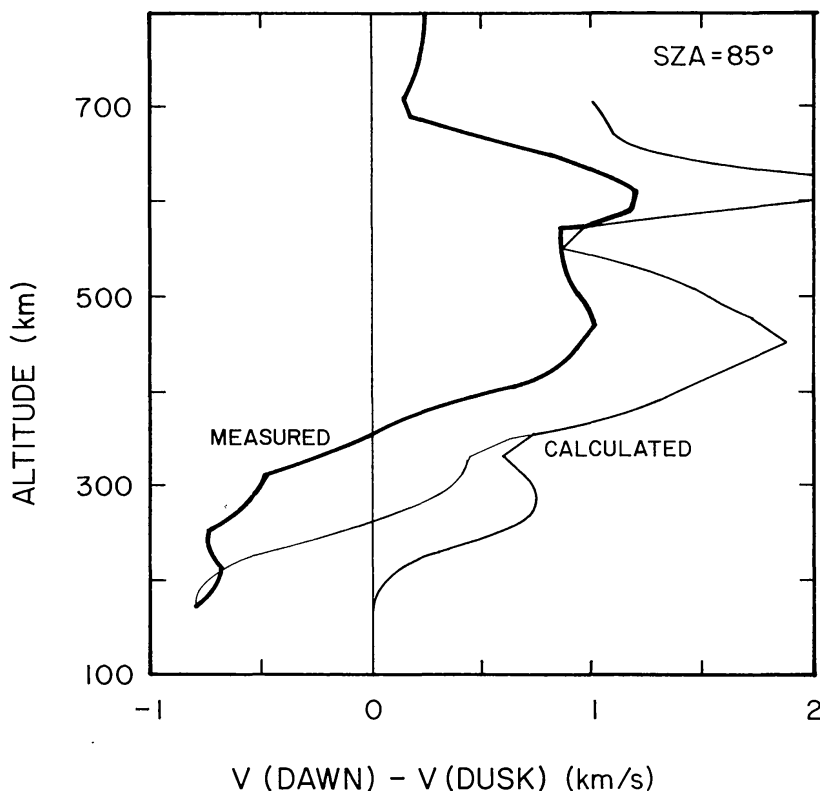


Fig. 16. Differences between average nightward velocity at the dawn and dusk terminators. The heavy line is the measured difference at 85° SZA. The light line is the calculated difference based on the momentum equation. The line joining the two curves below 400 km altitude is the calculated difference assuming a 400 m s^{-1} superrotation of the neutral atmosphere (after Miller and Knudsen, 1987).

dawn and dusk hemispheres at 85° SZA. The difference is negative below about 400 km, indicating superrotation, but positive above 400 km, which indicates a prograde rotation component. This is not actually a superrotation, but a dawn-dusk asymmetry in the in velocity.

It must be said here that the possibility that the superrotation component is a systematic measurement error has not been entirely ruled out. The ram direction of the spacecraft is toward the southwest during periapsis, and thus has a component in the direction of the measured superrotation. At the time of this writing, nothing has been found in the ORPA measurement that would result in an incorrect superrotation velocity component.

Since the internal magnetic field of Venus is negligible or non-existent, and induced fields are generally small, plasma flow is primarily in response to a gradient in the plasma pressure. As a first approximation, Elphic *et al.* (1984b) described the velocity using a simplified one-dimensional steady-state momentum equation coupled with measurements of ionospheric densities. Miller and Knudsen (1987) used ion concentration and temperature measurements by the ORPA (Miller *et al.*, 1984), together with the neutral model of Hedin *et al.* (1983) to evaluate this equation. The cumulative contribution of advection was approximated by beginning the calculation at the location of the median

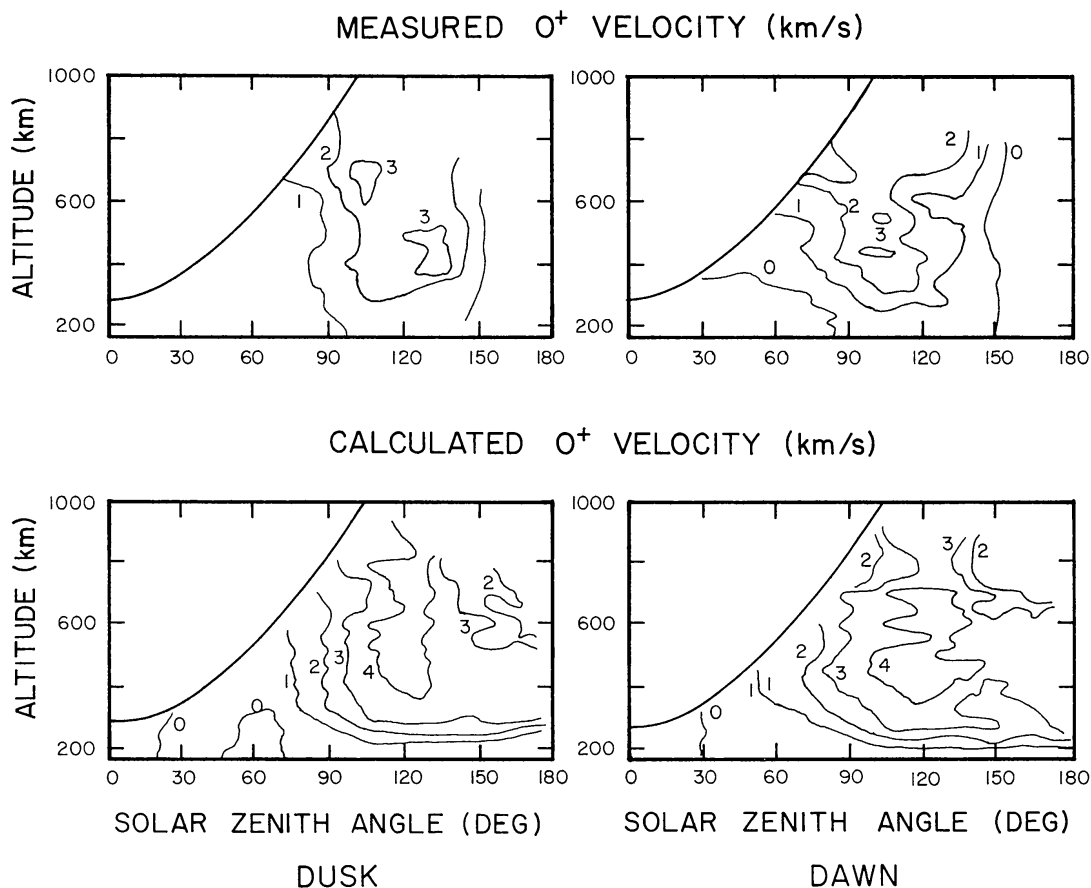


Fig. 17. Measured and calculated O^+ velocity in the dawn and dusk hemispheres. Calculations were based on a simplified momentum equation (after Miller and Knudsen, 1987).

ionopause (Knudsen *et al.*, 1982b). The velocity field based on the simplified momentum equation is compared with the measured average velocity in Figures 16 and 17.

The dayside results are consistent with measurements, both qualitatively and quantitatively. Contours in Figure 17 show a much greater vertical gradient in the velocity on the dawn hemisphere than on the dusk hemisphere. The calculated results for the nightside are probably inaccurate, because of, among other things, a strong downward flow behind the terminator and the measured end of ordered flow at about 140° SZA (Knudsen *et al.*, 1980a).

The difference in the computed velocities of the two hemispheres shows that the result of the momentum equation gives a prograde asymmetry that disappears at low altitude. The retrograde superrotation component at low altitude can be simulated by introducing a high-speed superrotation in the neutral atmosphere. The effect of collisions with the neutral atmosphere is dominant below 250 km, but decreases rapidly with altitude and becomes negligible above about 350 km. Figure 16 compares the measured dawn/dusk velocity difference at 85° SZA with the difference calculated with and without a neutral atmosphere that superrotates at a speed of 400 m s^{-1} .

The velocity differences at all SZA are compared with the calculated differences in Figure 18. It can be seen that the calculations agree quite well with the measurements.

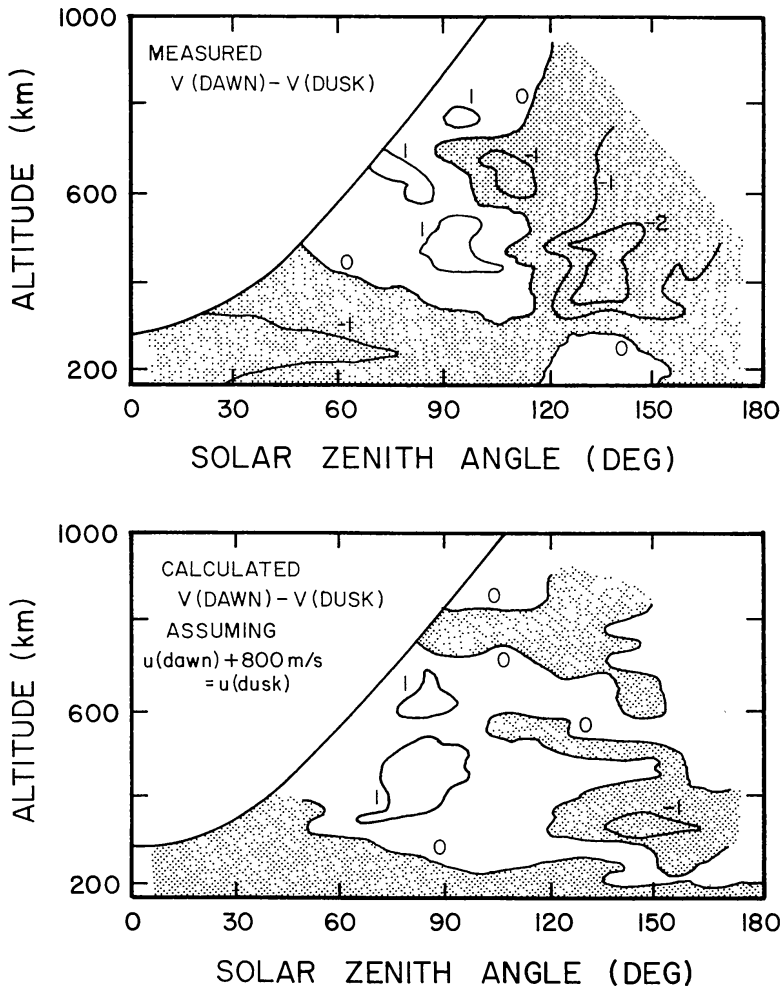
DAWN – DUSK O^+ VELOCITY DIFFERENCE (km/s)

Fig. 18. Measured and calculated velocity differences between dawn and dusk hemispheres. A super-rotation of the neutral atmosphere of 400 m s^{-1} has been included in the calculation (after Miller and Knudsen, 1987).

As before, calculated velocities on the nightside are probably in error because of influences on the velocity that are not directly related to the pressure gradient and not included in the model. For SZA less than approximately 110° , the calculation reproduces the general features shown in the measurement.

No one has suggested why the high-altitude ion velocities are greater at the dawn terminator than at the dusk terminator, although Miller and Knudsen (1987) showed that the difference is consistent with measured plasma pressure gradients. The asymmetry may be related to a similar dawn-dusk asymmetry in the ionopause height. Phillips *et al.* (1988) showed that the apparent asymmetry in ionopause height is a result of the combination of the aberration of the solar wind velocity due to the orbital motion of Venus and a measurement bias that is the result of the solar wind dynamic pressure being greater on the average during the time when the dusk ionosphere was sampled.

5.2. VARIABILITY IMPOSED BY FLUCTUATIONS IN THE SOLAR WIND

Experiments on Pioneer–Venus have shown large excursions in the height of the ionopause and in the local density, temperature, and velocity on a time-scale comparable to the 24-hour period of the Pioneer–Venus orbit. Ionospheric changes due to solar wind variations must certainly have a much shorter response time (i.e., Wolff *et al.*, 1982). Statistical studies, however, have shown the median ionosphere to be well-behaved, and to vary smoothly with solar zenith angle (Knudsen *et al.*, 1980a, 1981, 1982a, Miller *et al.*, 1980, 1984; Theis *et al.*, 1984). The range of variability in the dayside ionosphere is presented in the statistical treatment of the ionospheric parameters by Miller *et al.* (1984). Over the three Venus years while the Pioneer Venus periapsis was maintained in the ionosphere, about two-thirds of the measurements of ion density on the dayside were within a factor of two of the median. A similar variability is observed in the dayside ion temperature.

The variability in the nightside ionosphere is more than a factor of three greater than that on the dayside (Knudsen *et al.*, 1986). The change of variability from the dayside to the nightside of Venus is apparently caused by the differing ionization sources on the two hemispheres (Theis *et al.*, 1984). The dayside ionospheric variability is produced primarily by changes in solar EUV radiation (Taylor *et al.*, 1981a, b; Elphic *et al.*, 1984a), while changes in the nightside ionosphere are apparently the result of changes in the ion flux across the terminator (Knudsen *et al.*, 1980a; Cravens *et al.*, 1982).

Trans-terminator plasma flow can only be a significant source for the nocturnal ionosphere if the ionopause height is at least 300 km (Knudsen *et al.*, 1987). Should that height be lowered, as during periods of high solar wind dynamic pressure or low solar activity, the flow will shut off and another source such as suprathermal electrons must be responsible for any observed nighttime ionosphere (Spencer *et al.*, 1981; Cravens *et al.*, 1982; Knudsen *et al.*, 1987).

A simultaneous measurement of the nightside ion density and the flux of ionization across the terminator is not possible with the Pioneer–Venus orbit. Miller and Knudsen (1987) investigated a causal relationship between the ion density on the nightside and the ion flow past the terminator by correlating the variability of the ion density within the Venus nightside ionosphere with the variability of the height of the ionopause.

The relationship between nightside ion density and ionopause height was derived by normalizing the ion density measured on each specific orbit to the empirical model of Theis *et al.* (1984), and the corresponding ionopause height to the model of Knudsen *et al.* (1982b). There is essentially no correlation with ionopause height in the dayside ionosphere between 170 and 200 km altitude. On the nightside, however, the ion density and ionopause height are correlated. Figure 19 is a scatter plot of normalized ionopause heights vs normalized nightside ion density, illustrating this direct relationship.

Cravens *et al.* (1982) suggested that the flux of ions into the nightside is directly related to the height of the ionopause. The variability in the ionospheric density is thus directly correlated to variations in ion flux into the nightside. Changes in ionospheric pressure

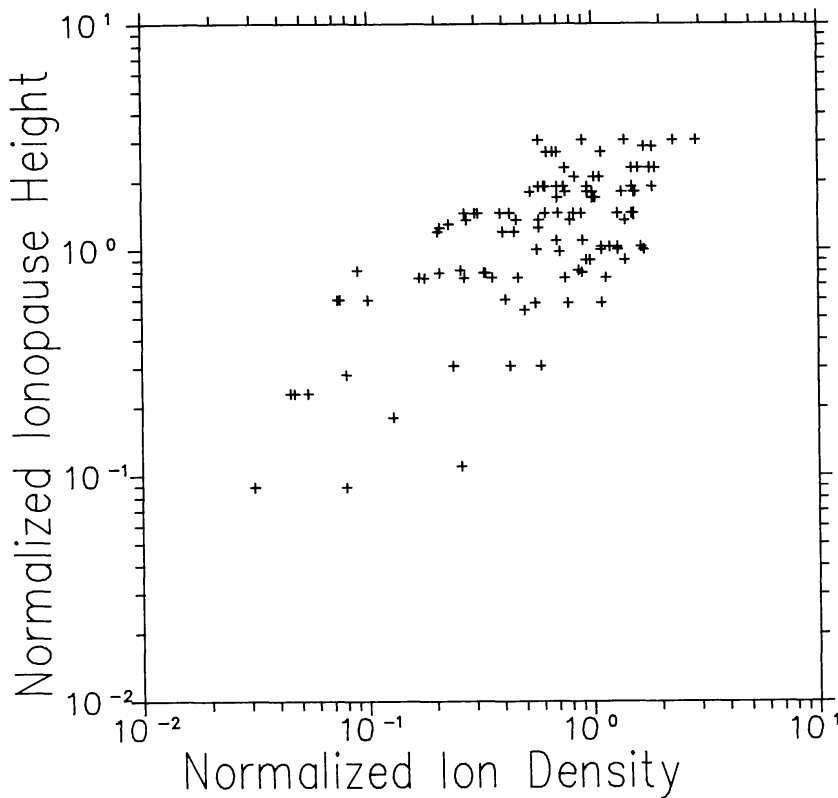


Fig. 19. Ionopause height plotted as a function of nightside ion density. Both quantities have been normalized by model calculations to allow for differences in location of the measurements (see text) (after Miller and Knudsen, 1987).

caused by variations in the solar EUV flux are not expected to be large from one orbit to the next, and changes in the solar wind pressure and the resulting changes in the height of the ionopause were suggested to cause the modulation of the nightside ionization by opening and closing the ion 'nozzle' at the terminator (Knudsen *et al.*, 1980a, 1981; Cravens *et al.*, 1982; Miller and Knudsen, 1987).

5.3. SOLAR CYCLE EFFECTS

Using six-month averages of solar wind data from Feldman *et al.* (1979), Knudsen *et al.* (1987) showed that the solar wind dynamic pressure does not vary significantly over a solar cycle, and, in fact, may be greater at solar minimum than at solar maximum. However, the solar EUV radiation has been observed to undergo very large variations, specifically by as much as a factor of about four (Oppenheimer *et al.*, 1981). As Knudsen *et al.* (1987, 1988) have shown, this variation in EUV irradiance very substantially reduces the ionospheric static pressure, mainly by decreasing the ionization rate of neutral atomic oxygen. As a result, the ionopause height is lowered until the solar wind dynamic pressure is balanced by the static pressure of the ionosphere or the solar wind is directly absorbed by the atmosphere of the planet (Luhmann *et al.*, 1987; Mahajan *et al.*, 1989). Because the dynamic pressure of the solar wind varies approximately as $\cos^2(\text{SZA})$, one can expect to find a balance between dynamic and static pressures as the terminator is approached from the dayside.

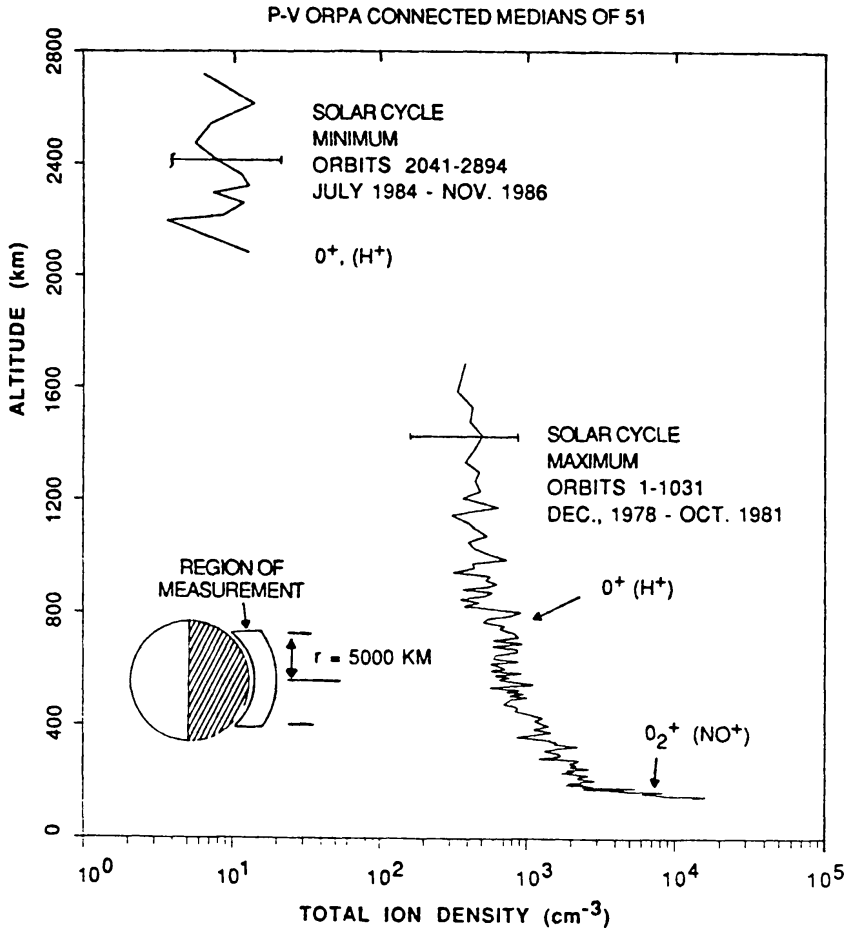


Fig. 20. Median ion densities on the nightside at solar maximum and at solar minimum. The discontinuity is interpreted as indicating the lack of O^+ transport from the dayside at solar minimum (after Knudsen, 1988).

Figure 20 shows median ion densities on the nightside of Venus measured by the ORPA. The data shown below 1600 km altitude are from the period between December 1978 and October 1981, while periapsis was being maintained in the lower ionosphere. This was also during the maximum of the solar cycle. The data shown at altitudes higher than 2000 km in Figure 20 are from the period between July 1984 and November 1986, near solar minimum. The periapsis altitude had been allowed to rise after 1981, accounting for the difference in altitude ranges of the two sets of data.

Knudsen *et al.* (1987) interpreted the difference between the nighttime ion densities of solar maximum and solar minimum as being a result of the lowering ionopause affecting the source of O^+ ions. The effect of the lowering of the ionopause at solar minimum is to effectively cut-off the ion flow from the dayside to the nightside. With the source of O^+ ions reduced, the dominant ionization source is impact ionization by low-energy suprathermal electrons (Gringauz *et al.*, 1979, 1983; Knudsen *et al.*, 1985). Thus the low-altitude molecular layer remains about the same, but there is a substantial reduction in the density of the high-altitude O^+ layer (Knudsen, 1988).

6. Summary

Ion velocities measured in the Venus ionosphere show the predominant flow to be away from the sub-solar point, and reaching speeds than 3 km s^{-1} in the terminator region. The axially-symmetric flow has been modelled using the laws of momentum, energy, and mass conservation. The modelling has generally been successful in reproducing the ion velocities at solar zenith angles less than about 140° . The chaotic flow and high ion temperature near the anti-solar point of the planet have not been successfully modelled. It has been suggested that the change from nightward flow to a chaotic flow near 140° SZA is the result of a recompression shock wave at the point where the ion velocity drops below the sonic velocity. The presence of a shock wave does not, however, explain the high ion temperatures that are observed.

Ion velocities are not strictly axially-symmetric, but, in a statistical sense, show a difference between the nightward flow in the dawn and dusk hemispheres. At altitudes below about 400 km, this asymmetry has the appearance of an ionospheric superrotation of about 400 m s^{-1} . There is also evidence of an ionospheric superrotation in the dawnward offset of the region of ion convergence on the nightside.

Above 400 km, the dawn-dusk asymmetry is in the opposite sense to the low-altitude superrotation component. Although no cause has been suggested for the greater nightward velocity at the dawn terminator, it has been shown to be consistent with a similar asymmetry in the plasma pressure gradients of the two hemispheres. The high-altitude asymmetry may, as is the case with a similar asymmetry in the ionopause height, be the result of differences in the solar wind and solar radiation at the times of the measurements.

Changes in the dynamic pressure of the solar wind have been shown to produce variability in the nightside ionosphere. During solar maximum, this variability is the result of transient increases in the solar wind pressure lowering the ionopause height and restricting the flow of O^+ ions to the nightside. The O^+ flux is restricted in a similar way during most of the solar minimum when the ionospheric pressure is reduced and allows the ionopause to remain near its lower limiting altitude.

Acknowledgements

This work was supported by NASA contracts NAGW1556 and NAGW869. The initial development of some sections of this manuscript was supported by Lockheed independent research funds.

References

- Banks, P. M. and Kockarts, G.: 1973, *Aeronomy*, Academic Press, New York.
- Bauer, S. J.: 1973, *Physics of Planetary Ionospheres*, Springer-Verlag, Berlin.
- Bougher, S. W. and Cravens, T. E.: 1984, 'Two-Dimensional Model of the Nightside Ionosphere of Venus: Ion Energetics', *J. Geophys. Res.* **89**, 3837.
- Brace, L. H. and Kliore, A. J.: 1991, 'The Structure of the Venus Ionosphere', *Space Sci. Rev.* **55**, 81.

- Braginskii, S.: 1958, 'Transport Phenomena in a Completely Ionized Two-Temperature Plasma', *Soviet Phys. JETP* **6**, 358.
- Chen, R. H.: 1977, 'The Venus Ionosphere', Ph.D. Thesis, University of Michigan, Ann Arbor.
- Chen, R. H. and Nagy, A. F.: 1978, 'A Comprehensive Model of the Venus Ionosphere', *J. Geophys. Res.* **83**, 1133.
- Cloutier, P. A.: 1984, 'Formation and Dynamics of Large-Scale Magnetic Structures in the Ionosphere of Venus', *J. Geophys. Res.* **89**, 2401.
- Cloutier, P. A., Tascione, T. F., and Daniell, R. E., Jr.: 1981, 'An Electrodynamic Model of Electric Currents and Magnetic Fields in the Dayside Ionosphere of Venus', *Planetary Space Sci.* **29**, 635.
- Cloutier, P. A., Tascione, T. F., Daniell, R. E., Jr., Taylor, H. A., Jr., and Wolff, R. S.: 1983, in D. M. Hunten, L. Colin, T. M. Donahue, and V. I. Moroz (eds.), 'Physics of the Interaction of the Solar Wind with the Ionosphere of Venus: Flow/Field Models', *Venus*, Univ. Arizona Press, Tucson, pp. 941–979.
- Cloutier, P. A., Taylor, H. A., Jr., and McGary, J. A.: 1987, 'Steady State Flow/Field Model of Solar Wind Interaction with Venus: Global Implications of Local Effects', *J. Geophys. Res.* **92**, 7289.
- Colin, L.: 1983, in D. M. Hunten, L. Colin, T. M. Donahue, and V. I. Moroz (eds.), 'Basic Facts about Venus', *Venus*, Univ. Arizona Press, Tucson, pp. 941–979.
- Cravens, T. E., Kliore, A. J., Kozyra, J. U., and Nagy, A. F.: 1981, 'The Ionospheric Peak on the Venus Dayside', *J. Geophys. Res.* **86**, 11323.
- Cravens, T. E., Brace, L. H., Taylor, H. A., Jr., Russell, C. T., Knudsen, W. C., Miller, K. L., Barns, A., Mihalof, J. D., Scarf, F. L., Quenon, S. J., and Nagy, A. F.: 1982, 'Disappearing Ionospheres on the Nightside of Venus', *Icarus* **51**, 271.
- Cravens, T. E., Crawford, S. L., Nagy, A. F., and Gombosi, T. I.: 1983, 'A Two-Dimensional Model of the Ionosphere of Venus', *J. Geophys. Res.* **88**, 5595.
- Cravens, T. E., Shinagawa, H., and Nagy, A. F.: 1984, 'The Evolution of Large-Scale Magnetic Fields in the Ionosphere of Venus', *Geophys. Res. Letters* **11**, 267.
- Elphic, R. C., Brace, L. H., Theis, R. F., and Russell, C. T.: 1984a, 'Venus Dayside Ionospheric Conditions: Effects of Ionospheric Magnetic Field and Solar EUV Flux', *Geophys. Res. Letters* **11**, 124.
- Elphic, R. C., Mayr, H. G., Theis, R. F., Brace, L. H., Miller, K. L., and Knudsen, W. C.: 1984b, 'Nightward Ion Flow in the Venus Ionosphere: Implications of Momentum Balance', *Geophys. Res. Letters* **11**, 1007.
- Feldman, W. C., Asbridge, J. R., Bame, S. J., and Gosling, J. T.: 1979, 'Long-Term Solar Wind Electron Variations between 1971 and 1978', *J. Geophys. Res.* **84**, 7371.
- Gombosi, T. I., Cravens, T. E., Nagy, A. F., Elphic, R. C., and Russell, C. T.: 1980, 'Solar Wind Absorption by Venus', *J. Geophys. Res.* **85**, 7747.
- Gringauz, K. I., Verigin, M. I., Breus, T. K., and Gombosi, T.: 1979, 'The Interaction of Electrons in the Optical Umbra of Venus with the Planetary Atmosphere – The Origin of the Nighttime Ionosphere', *J. Geophys. Res.* **84**, 2123.
- Gringauz, K. I., Verigin, M. I., Breus, T. K., and Shvachunova, L. A.: 1983, 'On the Prevailing Ionization Source in the Main Ionization Peak of the Nightside Ionosphere of Venus', *Kosmich Issled.* **21**, 746.
- Hedin, A. E., Niemann, H. B., Kasprzak, W. T., and Seiff, A.: 1983, 'Global Empirical Model of the Venus Thermosphere', *J. Geophys. Res.* **88**, 73.
- Heroux, L. and Hinteregger, H. E.: 1978, 'Aeronomical Reference Spectrum for Solar UV below 2000 Å', *J. Geophys. Res.* **83**, 5305.
- Hinteregger, H.: 1979, 'Development of Solar Cycle 21 Observed in EUV Spectrum and Atmospheric Absorption', *J. Geophys. Res.* **84**, 1933.
- Hinteregger, H.: 1981, 'Representations of Solar EUV Fluxes for Aeronomical Applications', *Adv. Space Res.* **1**, 39.
- Kar, J. and Mahajan, K. K.: 1987, 'On the Response of Ionospheric Magnetization to Solar Wind Dynamic Pressure from Pioneer Venus Measurements', *Geophys. Res. Letters* **14**, 507.
- Keating, G. M., Nicholson, J. Y., and Lake, L. R.: 1980, 'Venus Upper Atmosphere Structure', *J. Geophys. Res.* **85**, 7941.
- Knudsen, W. C.: 1988, 'Solar Cycle Changes in the Morphology of the Venus Ionosphere', *J. Geophys. Res.* **93**, 8756.
- Knudsen, W. C.: 1990, 'Role of Hot Oxygen in Venusian Ionospheric Ion Energetics and Supersonic Antisunward Flow', *J. Geophys. Res.* **95**, 1097.
- Knudsen, W. C., Miller, K. L., and Spenser, K.: 1985, 'Improved Venus Ionopause Altitude Calculation and Comparison with Measurement', *J. Geophys. Res.* **87**, 2246.

- Knudsen, W. C., Bakke, J. C., Spenner, K., and Novak, V.: 1979, 'Retarding Potential Analyzer for the Pioneer-Venus Orbiter Mission', *Space Sci. Instr.* **4**, 351.
- Knudsen, W. C., Spenner, K., Miller, K. L., and Novak, V.: 1980a, 'Transport of Ionospheric O^+ Ions Across the Venus Terminator and Implications', *J. Geophys. Res.* **85**, 7803.
- Knudsen, W. C., Spenner, K., Whitten, R. C., and Miller, K. L.: 1980b, 'Ion Energetics in the Venus Nightside Ionosphere', *Geophys. Res. Letters* **7**, 1045.
- Knudsen, W. C., Spenner, K., Bakke, J., and Novak, V.: 1980c, 'Pioneer Venus Orbiter Palanar Retarding Potential Analyzer Plasma Experiment', *IEEE Transactions on Geoscience and Remote Sensing* **GE-18**, 54.
- Knudsen, W. C., Spenner, K., and Miller, K. L.: 1981, 'Anti-Solar Acceleration of Ionospheric Plasma Across the Venus Terminator', *Geophys. Res. Letters* **8**, 241.
- Knudsen, W. C., Banks, P. M., and Miller, K. L.: 1982a, 'A Model of Plasma Motion and Planetary Magnetic Fields for Venus', *Geophys. Res. Letters* **9**, 765.
- Knudsen, W. C., Miller, K. L., and Spenner, K.: 1982b, 'Improved Venus Ionopause Altitude Calculation and Comparison with Measurement', *J. Geophys. Res.* **87**, 2246.
- Knudsen, W. C., Miller, K. L., and Spenner, K.: 1986, 'Median Density Altitude Profiles of the Major Ions in the Central Nightside Venus Ionosphere', *J. Geophys. Res.* **91**, 11936.
- Knudsen, W. C., Kliore, A. J., and Whitten, R. C.: 1987, 'Solar Cycle Changes in the Ionization Sources of the Nightside Venus Ionosphere', *J. Geophys. Res.* **92**, 13391.
- Luhmann, J. G. and Cravens, T. E.: 1991, 'Magnetic Fields in the Ionosphere of Venus', *Space Sci. Rev.* **55**, 201.
- Luhmann, J. G., Russell, C. T., and Elphic, R. C.: 1984, 'Time-Scales for the Decay of Induced Large-Scale Magnetic Fields in the Venus Ionosphere', *J. Geophys. Res.* **89**, 362.
- Luhmann, J. G., Russell, C. T., Scarf, F. L., Brace, L. H., and Knudsen, W. C.: 1987, 'Characteristics of the Marslike Limit of the Venus-Solar Wind Interaction', *J. Geophys. Res.* **92**, 8545.
- Mahajan, K. K. and Kar, J.: 1988, 'Planetary Ionospheres', *Space Sci. Rev.* **47**, 303.
- Mahajan, K. K., Mayr, H. G., Brace, L. H., and Cloutier, P. A.: 1989, 'On the Lower Altitude Limit of the Venusian Ionopause', *Geophys. Res. Letters* **16**, 759.
- Mayr, H. G., Harris, I., Niemann, H. B., Brinton, H. C., Spencer, N. W., Taylor, H. A., Hartle, R. E., Hoegy, W. R., and Hunten, D. M.: 'Dynamic Properties of the Thermosphere Inferred from Pioneer Venus Mass Spectrometer Measurements', *J. Geophys. Res.* **85**, 7841.
- Mayr, H. G., Harris, I., Stevens-Rayburn, D. R., Niemann, H. B., Taylor, H. A., Jr., and Hartle, R. E.: 1985, 'On the Diurnal Variations in the Temperature and Composition: A Three-Dimensional Model with Superrotation', *Adv. Space Res.* **5**, 109.
- McCormick, P. T., Whitten, R. C., and Knudsen, W. C.: 1987, 'Dynamics of the Venus Ionosphere Revisited', *Icarus* **70**, 469.
- Merritt, D. and Thompson, K.: 1980, 'Thermal Energy Transport in the Venus Ionosphere: Classical and Saturated Electron Temperature Profiles', *J. Geophys. Res.* **85**, 6778.
- Miller, K. L. and Knudsen, W. C.: 1987, 'Spatial and Temporal Variations of the Ion Velocity Measured in the Venus Ionosphere', *Adv. Space Res.* **7**(12), 107.
- Miller, K. L., Knudsen, W. C., Spenner, K., Whitten, R. C., and Novak, V.: 1980, 'Solar Zenith Angle Dependence of Ionospheric Ion and Electron Temperatures and Density on Venus', *J. Geophys. Res.* **85**, 7759.
- Miller, K. L., Knudsen, W. C., and Spenner, K.: 1984, 'The Dayside Venus Ionosphere. I. Pioneer-Venus Retarding Potential Analyzer Experimental Observations', *Icarus* **57**, 386.
- Nagy, A. F. and Cravens, T. E.: 1988, 'Hot Oxygen Atoms in the Atmospheres of Venus and Mars', *Geophys. Res. Letters* **15**, 433.
- Nagy, A. F., Cravens, T. E., Smith, S. G., Taylor, H. A., Jr., and Brinton, H. C.: 1980, 'Model Calculations of the Dayside Ionosphere of Venus: Ionic Composition', *J. Geophys. Res.* **85**, 7795.
- Nagy, A. F., Cravens, T. E., Lee, J. H., and Stewart, A. E. F.: 1981, 'Hot Oxygen Atoms in the Upper Atmosphere of Venus', *Geophys. Res. Letters* **8**, 629.
- Nakada, M. P. and Sullivan, E. C.: 1980, 'Thermal Diffusion Calculations for the Ionosphere of Venus', *J. Geophys. Res.* **85**, 171.
- Niemann, H. B., Kasprzak, W. T., Hedin, A. E., Hunten, D. M., and Spencer, N. W.: 1980, 'Mass Spectrometric Measurements of the Neutral Gas Composition of the Thermosphere and Exosphere of Venus', *J. Geophys. Res.* **85**, 7817.

- Oppenheimer, M., Babeu, S., and Brinton, H. C.: 1981, 'EUV Flux Variations During Solar Cycle 21 from AE-E He⁺ Abundances', *J. Geophys. Res.* **86**, 825.
- Paxton, L. J. and Stewart, A. E.: 1988, 'The Hot Oxygen Corona of Venus', *J. Geophys. Res.* (submitted).
- Perez-de-Tejada, H.: 1982, 'Viscous Dissipation at the Venus Ionopause', *J. Geophys. Res.* **87**, 7405.
- Phillips, J. L., Luhmann, J. G., Knudsen, W. C., and Brace, L. H.: 1988, 'Asymmetries in the Location of the Venus Ionopause', *J. Geophys. Res.* **93**, 3927.
- Rohrbaugh, R. P., Nisbet, J. S., Bleuler, E., and Herman, J. R.: 1979, 'The Effect of Energetically Produced O₂⁺ on the Ion Temperatures of the Martian Thermosphere', *J. Geophys. Res.* **84**, 3327.
- Russell, C. T., Elphic, R. C., and Slavin, J. A.: 1980, 'Limits on the Possible Intrinsic Magnetic Field of Venus', *J. Geophys. Res.* **85**, 8319.
- Russell, C. T., Luhmann, J. G., and Elphic, R. C.: 1983, 'The Properties of the Low Altitude Magnetic Belt in the Venus Ionosphere', *Adv. Space Res.* **2**, 13.
- Schubert, G.: 1983, in D. M. Hunten, L. Colin, T. M. Donahue, and V. I. Moroz (eds.), 'General Circulation and the Dynamical State of the Venus Atmosphere', *Venus*, Univ. Arizona Press, Tucson, pp. 681–765.
- Schunk, R. and Walker, J. C. G.: 1969, 'Thermal Diffusion in the Topside Ionosphere for Mixtures which Include Multiply-Charged Ions', *Planetary Space Sci.* **17**, 853.
- Shinagawa, H. and Cravens, T. E.: 1988, 'A One-Dimensional Multispecies Magnetohydrodynamic Model of the Dayside Ionosphere of Venus', *J. Geophys. Res.* **93**, 11263.
- Shinagawa, H., Cravens, T. E., and Nagy, A. F.: 1987, 'A One-Dimensional Time-Dependent Model of the Magnetized Ionosphere of Venus', *J. Geophys. Res.* **92**, 7317.
- Singhal, R. P. and Whitten, R. C.: 1986, 'A Two-Dimensional Model of the Ionosphere of Venus: Thermal Structure', *Icarus* **67**, 325.
- Singhal, R. P. and Whitten, R. C.: 1987, 'A Simple Spectral Model of the Dynamics of the Venus Ionosphere', *J. Geophys. Res.* **92**, 5735.
- Singhal, R. P. and Whitten, R. C.: 1988, 'Horizontal Plasma Flow Velocities in the Ionosphere of Mars: A Test Case for the Solar Wind Interaction', *Ind. J. Radio Space Res.* **17**, 42.
- Spencer, K., Knudsen, W. C., Whitten, R. C., Michelson, P. F., Miller, K. L., and Novak, V.: 1981, 'On the Maintenance of the Venus Nighttime Ionosphere: Electron Precipitation and Plasma Transport', *J. Geophys. Res.* **86**, 9170.
- Taylor, H. A., Jr., Brinton, H. C., Bauer, S. J., Hartle, R. E., Cloutier, P. A., Michel, F. C., Daniell, R. E., Donahue, T. M., and Maehl, R. C.: 1979, 'Ionosphere of Venus: First Observations of the Effects of Dynamics on the Dayside Ion Composition', *Science* **203**, 755.
- Taylor, H. A., Jr., Brinton, H. C., Wagner, T. C. G., Blackwell, B. H., and Cordier, G. R.: 1980a, 'Bennett Ion Mass Spectrometers on the Pioneer Venus Bus and Orbiter', *IEEE Transactions on Geoscience and Remote Sensing* **GE-18**, 44.
- Taylor, H. A., Jr., Brinton, H. C., Bauer, S. J., Hartle, R. E., Cloutier, P. A., and Daniell, R. E.: 1980b, 'Global Observations of the Composition and Dynamics of the Ionosphere of Venus: Implications for the Solar Wind Interaction', *J. Geophys. Res.* **85**, 7765.
- Taylor, H. A., Jr., Bauer, S. J., Daniell, R. E., Brinton, H. C., Mayr, H. E., and Hartle, R. E.: 1981a, 'Temporal and Spatial Variations Observed in the Ionospheric Composition of Venus – Implications for Empirical Modelling', *Adv. Space Res.* **1**, 37.
- Taylor, H. A., Jr., Daniell, R. E., Hartle, R. E., Brinton, H. C., Bauer, S. J., and Scarf, F. L.: 'Dynamic Variations Observed in Thermal and Superthermal Ion Distributions in the Dayside Ionosphere of Venus', *Adv. Space Res.* **1**, 247.
- Theis, R. F., Brace, L. H., Elphic, R. C., and Mayr, H. G.: 1984, 'New Empirical Models of the Electron Temperature and Density in the Venus Ionosphere with Application to Transterminator Flow', *J. Geophys. Res.* **89**, 1477.
- Von Zahn, U., Fricke, K. H., Hunten, D. M., Krankowsky, D., Mauersberger, K., and Nier, A. O.: 1980, 'The Upper Atmosphere of Venus During Morning Conditions', *J. Geophys. Res.* **85**, 7829.
- Whitten, R. C.: 1969, 'Thermal Structure of the Ionosphere of Venus', *J. Geophys. Res.* **74**, 5623.
- Whitten, R. C. and Knudsen, W. C.: 1980, 'Simple Models of the Thermal Structure of the Venusian Ionosphere', *Icarus* **44**, 85.
- Whitten, R. C., Baldwin, B., Knudsen, W. C., Miller, K. L., and Spencer, K.: 1982, 'The Venus Ionosphere at Grazing Incidence of Solar Radiation: Transport of Plasma to and in the Nightside', *Icarus* **51**, 261.
- Whitten, R. C., McCormick, P. T., Merritt, D., Thompson, K. W., Brynsvold, R. R., Eich, C. J., Knudsen,

1991SSRV...55..165M

- W. C., and Miller, K. L.: 1984, 'Dynamics of the Venus Ionosphere: A Two-Dimensional Model Study', *Icarus* **60**, 317.
- Whitten, R. C., Singhal, R. P., and Knudsen, W. C.: 1986, 'Thermal Structure of the Venus Ionosphere: A Two-Dimensional Model Study', *Geophys. Res. Letters* **13**, 10.
- Wolff, R. S., Stein, R. F., and Taylor, H. A., Jr.: 1982, 'The Dynamics of the Venus Ionosphere. 1. A Simulation of the Solar Wind Compression of the Upper Dayside Ionosphere', *J. Geophys. Res.* **87**, 8118.